

Design of Complex Integrated Circuits

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2026-09-11

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1 Introduction

Integrated circuits (ICs) are ubiquitous in our modern society. The extremely high density of functions (computation, storage, signal processing, digitization, energy conversion) offered by microelectronics enables all the devices and infrastructure that we depend on: the control of factories and energy generation, computation in server farms, personal computers and mobile devices, communication by wires, optics, or wireless links, and sensing of the environment. All of these are driven by integrated circuits, implemented in various semiconductor technologies.

This course looks at integrated circuits from two complementary angles. In the first part, we discuss how ICs are **built**: the operation of the MOSFET, the CMOS manufacturing technology, the physical layout, packaging, production testing, and the economics of chip making. In the second part, we discuss how complex ICs are **designed**: design methodologies for digital and mixed-signal circuits, protection against electrostatic discharge and latch-up, robust design techniques, and the inner workings of circuit simulators.

It is assumed that readers are familiar with the contents of our Analog Circuit Design course; basic knowledge of digital and analog circuit design is a prerequisite for the later chapters.

! Important

All course material (source code of this document, Python code for all figures, etc.) is made publicly available on GitHub and shared under the Apache-2.0 license.

Please feel free to submit pull requests to fix typos or add content! If you want to discuss something that is not clear, please open an issue.

The production of this document would be impossible without these (and many more) great open-source software products: VS Code, Quarto, Pandoc, LaTeX, Jupyter Notebook, Python, schemdraw, Numpy, Matplotlib, Pandas, Git, ...

1.1 How Small Is a Nanometer?

Modern CMOS technologies pattern structures with dimensions of a few nanometers. To develop a feeling for how small these dimensions really are, Figure 1 compares the feature sizes of integrated circuits with a few familiar objects: a human hair is roughly $70\text{ }\mu\text{m}$ thick, a red blood cell has a diameter of about $8\text{ }\mu\text{m}$, a virus measures around 150 nm , and the distance between two silicon atoms in the crystal lattice is only 0.24 nm . The minimum feature sizes of leading-edge CMOS processes are thus much smaller than a virus, and only about one order of magnitude away from the atomic spacing of the silicon crystal itself!

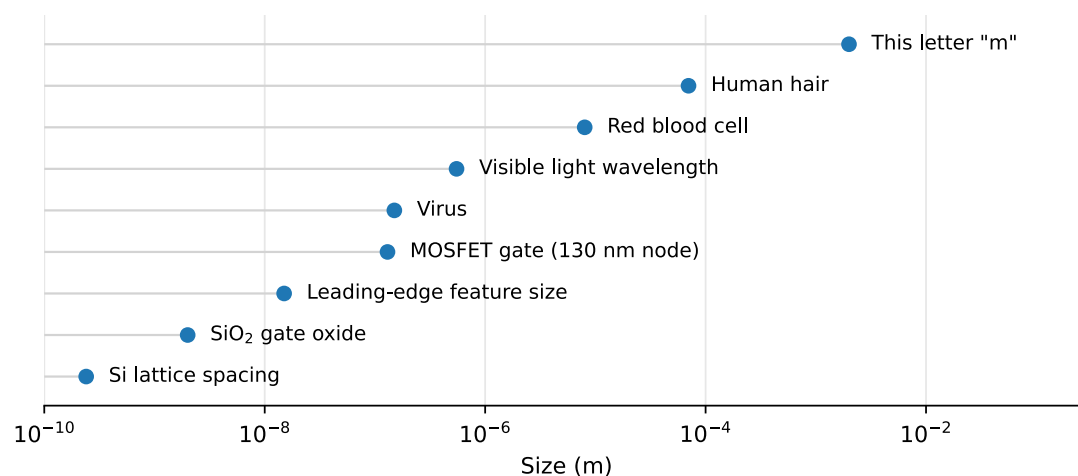


Figure 1: Length scales of integrated-circuit structures compared to familiar objects, on a logarithmic axis.

1.2 A Brief History of Semiconductors

The history of semiconductor electronics spans not even one century from the first observation of semiconductor rectification to today's chips with billions of transistors. Figure 2 summarizes the key milestones which we briefly discuss in the following.

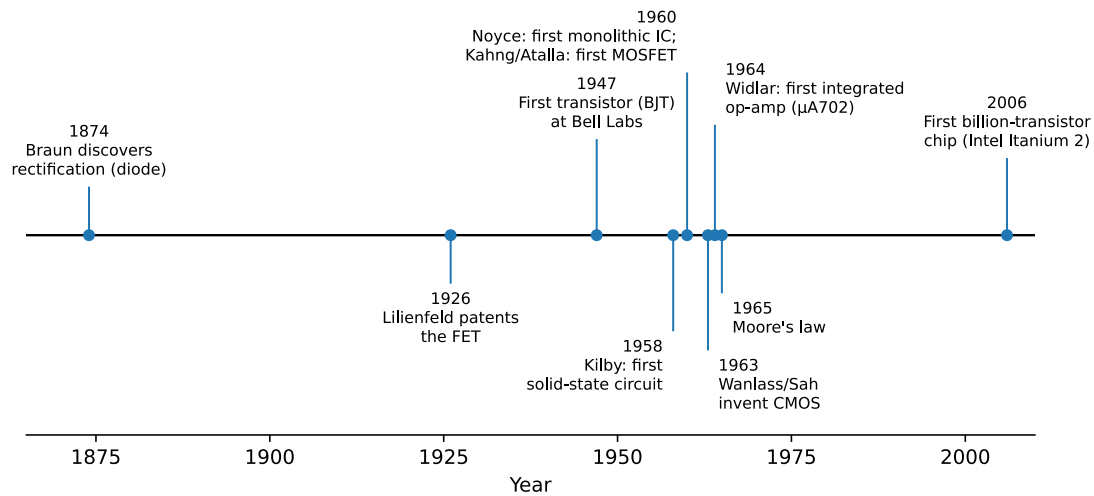


Figure 2: Timeline of key milestones in semiconductor history.

1874, the diode: Karl Ferdinand Braun discovers the **rectifying effect** of a particular type of mineral (galena, “Bleiglanz”). This effect is later exploited in radio receivers in the form of the crystal detector (the “cat’s whisker”), which is the first semiconductor electronic device.

1926, the field-effect transistor (FET): Julius Lilienfeld patents the **field-effect transistor**. Unfortunately, it takes several decades before such a device can actually be manufactured, as surface charges at the semiconductor interface cause malfunction of the early devices. Only in 1960 do Dawon Kahng and Mohamed Atalla finally demonstrate the metal-oxide-semiconductor field-effect transistor (MOSFET).

1947, the bipolar junction transistor (BJT): John Bardeen, William Shockley, and Walter Brattain demonstrate the first **semiconductor transistor** (using germanium) at Bell Labs. This was a big step towards the replacement of vacuum tubes. In the first point-contact transistor, the bottom piece of germanium formed the base terminal, hence the name “base.”

1958/1960, the first integrated circuit: After the first solid-state circuit demonstrated by Jack Kilby in 1958, Robert Noyce and his team at Fairchild build the first **monolithic integrated circuit** with pn-junction isolation in 1960: a flip-flop in transistor-resistor logic, integrated onto a wafer with 3 mm diameter.

1963, complementary MOS (CMOS): In an ISSCC paper, Frank Wanlass and Chih-Tang Sah of Fairchild Semiconductor show that logic built from p-channel and n-channel MOSFETs draws very little standby power—the invention of CMOS [1].

1964, analog integrated circuits: Robert J. Widlar, the “Father of Analog Integrated Circuits,” designs the first integrated operational amplifier, the μ A702.

1965, Moore’s law: Gordon E. Moore, based on only a handful of data points, makes a bold prediction about the growth of the number of components per integrated circuit, which would eventually become known as “**Moore’s law**” [2]. Figure 3 shows the data available to Moore in 1965 together with his famous extrapolation: a doubling of the number of components per chip every year (later revised to a doubling every two years)—an exponential growth trend that the industry has followed for decades.

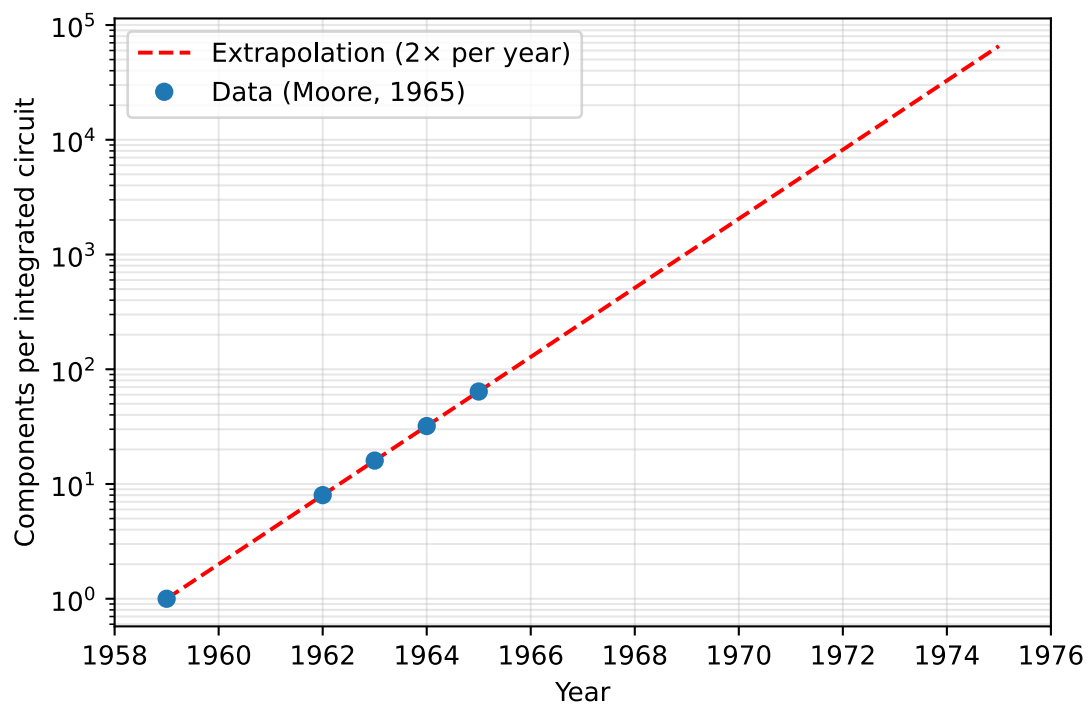


Figure 3: Moore's law: the number of components per integrated circuit versus time. The blue points reproduce the data available to Moore in 1965; the dashed red line is his extrapolation of a doubling every year (data after Moore, 'Cramming more components onto integrated circuits,' Electronics, 1965).

2006, the first billion-transistor logic chip: Intel announces the first microprocessor with more than one billion transistors, the Intel Itanium 2. The chip is manufactured in a 90 nm CMOS process and contains 1.72 billion transistors on a die with an area of 374 mm². The first demonstrated chip with one billion transistors is a 1 Gbit DRAM presented by NEC in 1995, manufactured in 0.25 μ m technology [3].

For an excellent, much more detailed account of semiconductor history, visit the Silicon Engine timeline of the Computer History Museum.

2 The MOSFET

This chapter on the metal-oxide-semiconductor field-effect transistor (MOSFET) is largely based on [4], [5], [6], which are also recommended as further reading. In contrast to our Analog Circuit Design course, where we treated the MOSFET essentially as a numerically-characterized black box, we will derive a simple, first-order model of the MOSFET from the physics of the device, and then discuss the most important second-order effects. The goal is to understand the basic operation of the MOSFET and its I/V characteristic, which will be used in later chapters for circuit design.

2.1 MOSFET Operation in a Nutshell

The basic MOSFET structure consists of a gate (terminal G) that is insulated from a doped semiconductor and controls the surface charge in the semiconductor channel below, between the source (terminal S) and the drain (terminal D). The source provides the charge carriers, and an electric field pulls them toward the drain (hence a semiconductor device based on drift

current). The electric potential of the semiconductor bulk (terminal B) acts on the channel as well and is the fourth terminal of the MOSFET. Ideally, the effect of the bulk is negligible.

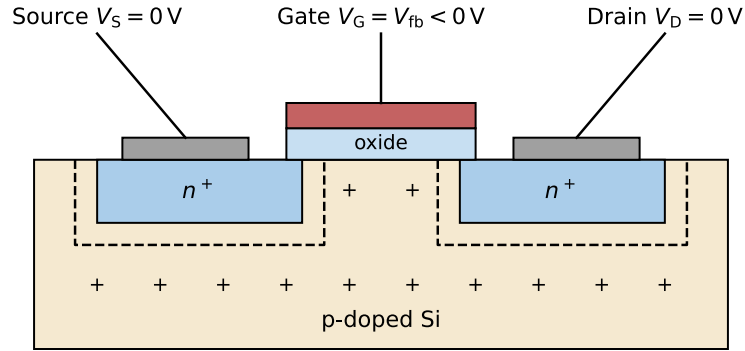
Two complementary flavors of the device exist: the n -type MOSFET (“NMOS”) has n^+ -doped source/drain regions on a p -type substrate and usually an n^{++} -doped gate, while the p -type MOSFET (“PMOS”) has p^+ -doped source/drain regions in an n -type well and usually a p^{++} -doped gate.

i Doping Notation

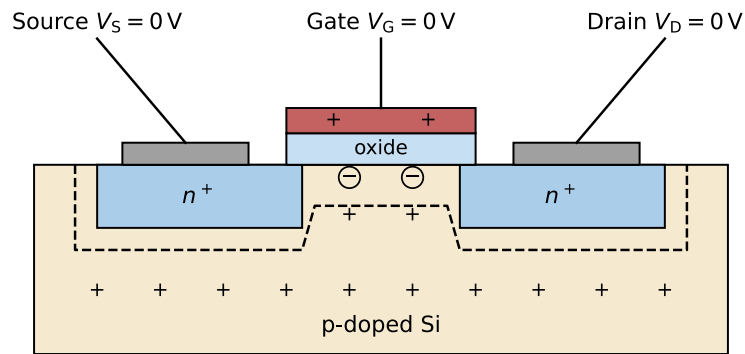
In the doping notation, $+$ indicates a higher doping concentration than the surrounding material, while $++$ indicates an even higher doping concentration. For example, n^+ -doped regions have a higher electron concentration than the n -type substrate, and n^{++} -doped gates have an even higher electron concentration than the n^+ -doped source/drain regions.

To understand the operation of the NMOS transistor, it is instructive to follow the conditions in the semiconductor while the gate voltage is swept upwards, as illustrated in Figure 4:

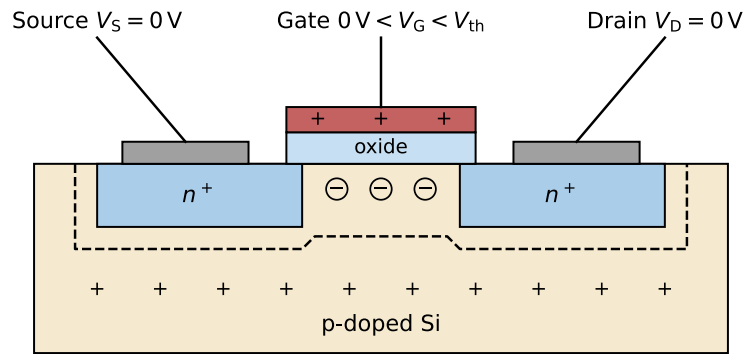
(a)



(b)



(c)



(d)

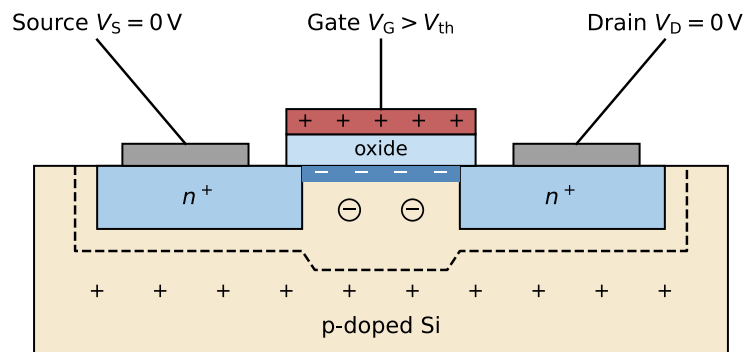
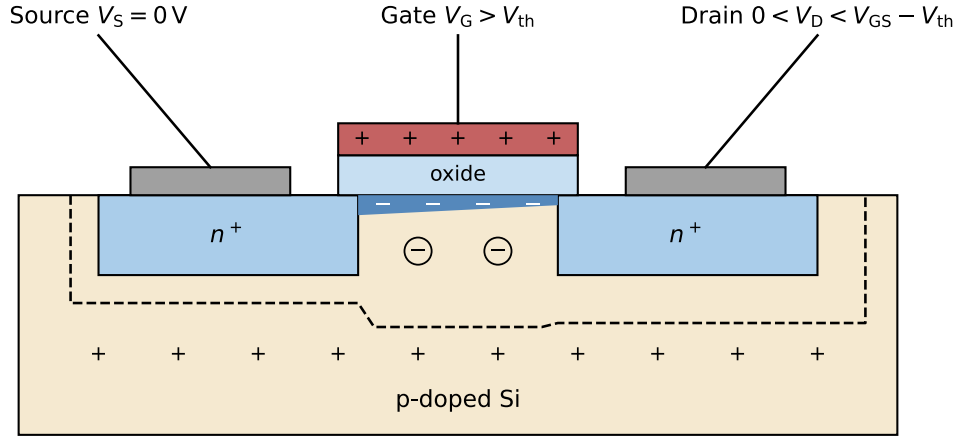


Figure 4: Cross section of an NMOS transistor for increasing gate voltage (drain and source grounded): (a) flatband condition, (b) gate at 0 V , (c) gate below the threshold voltage, and (d) inversion. The dashed line marks the depletion-region boundary, circled minus signs are ionized acceptors, plus signs are holes, and the dark blue layer in (d) is the electron inversion layer.

- **Flatband (a):** When the gate is at a specific negative voltage (the flatband voltage V_{fb}), and all other terminals are at 0 V, a depletion zone forms only around the drain and source junctions. No current can flow, since drain and source are n^+ -doped while the substrate is p -type, so the two junctions form back-to-back diodes.
- **Zero gate voltage (b):** When the gate voltage is increased to 0 V, a depletion zone also forms below the gate. The p -type acceptor ions in this region have captured an electron and are thus immobile, negatively charged sites (illustrated by the circled minus symbols)—still, no current can flow.
- **Below threshold (c):** When the gate voltage increases further (but stays below the threshold voltage), the depletion zone below the gate deepens, reaching a maximum depletion width W_{dep} when $V_{GS} = V_{th}$. There is still no conduction between drain and source, because there are no (or at least very few) free electrons in the channel region.
- **Inversion (d):** When the gate voltage rises above the threshold voltage V_{th} , **inversion** occurs: a thin layer of free electrons forms directly below the gate oxide, creating a conducting channel of electrons between drain and source. The source of the electrons in the inversion layer is the n^+ -doped source region, which is connected to ground. The electrons are pulled from the source into the channel by the positive gate voltage, and they can now flow to the drain if a drain voltage is applied.

Once the channel exists, the drain voltage determines the mode of conduction, as shown in Figure 5:

(a)



(b)

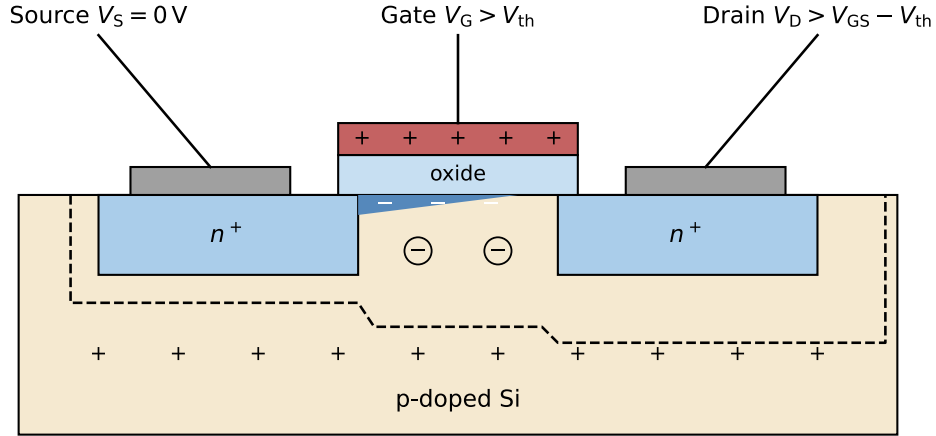


Figure 5: Cross section of an NMOS transistor in inversion for increasing drain voltage: (a) triode region, where the inversion layer extends from source to drain, and (b) saturation, where the channel is pinched off at the drain side.

- **Triode (a):** If a small voltage is applied between drain and source, a current flows, and the depletion zone around the drain widens. The drain current is a function of both V_{GS} and V_{DS} ; the device operates in the **triode** region. The MOSFET operates like a *voltage-controlled resistor* in this region.
- **Saturation (b):** If the drain-source voltage is increased beyond $V_{DS} > V_{GS} - V_{th}$, the channel is pinched off at the drain side, and the drain current is (to first order) no longer a function of V_{DS} , only of V_{GS} —the device is now in **saturation**. The MOSFET operates like a *voltage-controlled current source* in this region.

2.1.1 An Exemplary 130 nm Process Technology

Throughout this text we will use typical parameters of a 0.13 μm CMOS process technology, specifically IHP SG13 (process specification), for exemplary calculations. The key parameters are listed in Table 1.

Table 1: Typical parameters of the IHP SG13 130 nm CMOS technology

Parameter	Symbol	Typical value
Nominal supply voltage	V_{DD}	1.2 V (digital) / 1.5 V (analog)
Minimum gate length	L_{min}	130 nm
Oxide thickness	t_{ox}	2 nm
Threshold voltage NMOS PMOS	$V_{th,n} V_{th,p}$	0.5 V -0.47 V
Maximum voltage	$ V_{GS,max} , V_{DS,max} $	2 V
Gate capacitance	C'_{ox}	17.3 fF/ μm^2
Process constant $K_n K_p$	$\mu_n C'_{ox} \mu_p C'_{ox}$	280 $\mu A/V^2$ 130 $\mu A/V^2$

2.2 Basic I/V Characteristic of the MOSFET

2.2.1 The MOS Capacitor and the Threshold Voltage

If source, drain, and substrate are grounded ($V_S = V_D = V_B = 0$ V), and a positive voltage $V_{GS} = V_G$ is applied to the gate, then positive charge accumulates on the gate and negative charge in the p -type substrate, forming a depletion layer in the substrate.

If we assume a uniform charge density $\rho = -qN_a$ across this layer with N_a being the doping concentration of the p -type substrate (in a practical device, this charge density is not uniform), then we can use Poisson's equation. Note that with the boundary conditions assuming $\Phi(0) = 0$ and $d\Phi(0)/dy = 0$, $y = 0$ is the point where the depletion layer starts in the depth of the substrate, and $y = W_{dep}$ is the point where the depletion layer ends (the depletion width) at the substrate-oxide interface (ϵ_{si} is the permittivity of silicon):

$$\frac{d^2\Phi(y)}{dy^2} = -\frac{\rho(y)}{\epsilon_{si}} = \frac{qN_a}{\epsilon_{si}} \rightarrow \Phi(y) = \frac{qN_a}{2\epsilon_{si}}y^2 \quad (1)$$

The surface potential $\Phi_S = \Phi(W_{dep})$ at the oxide-silicon interface is then related to the depletion width W_{dep} by

$$W_{dep} = \sqrt{\frac{2\epsilon_{si}\Phi_S}{qN_a}}. \quad (2)$$

With W_{dep} we can also calculate the corresponding depletion capacitance (per unit area, indicated with the prime):

$$C'_{dep} = \frac{\epsilon_{si}}{W_{dep}} = \sqrt{\frac{qN_a\epsilon_{si}}{2\Phi_S}} \quad (3)$$

The charge per unit area Q'_{dep} in this region is (negative, because of the captured electrons)

$$Q'_{dep} = -qN_a W_{dep} = -\sqrt{2qN_a\epsilon_{si}\Phi_S} \quad (4)$$

The gate voltage is the result of the sum of the flatband voltage, the surface potential, and the voltage across the oxide

$$V_G = \Phi_{fb} + \Phi_S + V_{ox} = \Phi_{fb0} + \frac{-Q'_{ss}}{C'_{ox}} + \Phi_S + \frac{-Q'_{sub}}{C'_{ox}} \quad (5)$$

where Φ_{fb} is the work-function difference between the gate and the substrate material (the flatband voltage V_{fb} used in Figure 4). Note that the negative sign in the last term arises because voltage and charge are taken from different electrodes of the gate capacitor: the voltage V_{ox} at the top, and the substrate charge Q'_{sub} as well as the interface charge Q'_{ss} at the bottom.

An interface charge Q'_{ss} can be present at the SiO_2 -Si interface due to crystal discontinuities. This interface charge can be accounted for by adapting the flatband voltage Φ_{fb0} to Φ_{fb} (going forward, we will just use Φ_{fb}).

Using Equation 2 to express $\Phi_S = qN_a W_{dep}^2 / (2\varepsilon_{si})$ and Equation 4 for $Q'_{sub} = Q'_{dep}$, Equation 5 can be written as

$$V_G = \Phi_{fb} + \frac{qN_a W_{dep}^2}{2\varepsilon_{si}} + \frac{qN_a W_{dep}}{C'_{ox}} \quad (6)$$

From Equation 6, the depletion width W_{dep} can be calculated as a function of V_G , and from this $\Phi_S = f(V_G)$ using Equation 2.

If the surface potential Φ_S reaches a critical level of twice the Fermi potential Φ_B , then **inversion** occurs: a thin layer of electrons is induced directly under the oxide. The gate voltage that is minimally required to produce this inversion layer is called the **threshold voltage** $V_{th} = V_G(\Phi_S = 2\Phi_B)$.

$$V_{th} = \Phi_{fb} + 2\Phi_B + \frac{-Q'_{dep}}{C'_{ox}} = \Phi_{fb} + 2\Phi_B + \frac{\sqrt{2qN_a\varepsilon_{si}} \cdot 2\Phi_B}{C'_{ox}} \quad (7)$$

with the Fermi potential

$$\Phi_B = \frac{kT}{q} \ln \frac{N_a}{n_i}. \quad (8)$$

Figure 6 illustrates the flatband and the threshold condition in terms of the energy bands of the MOS structure.

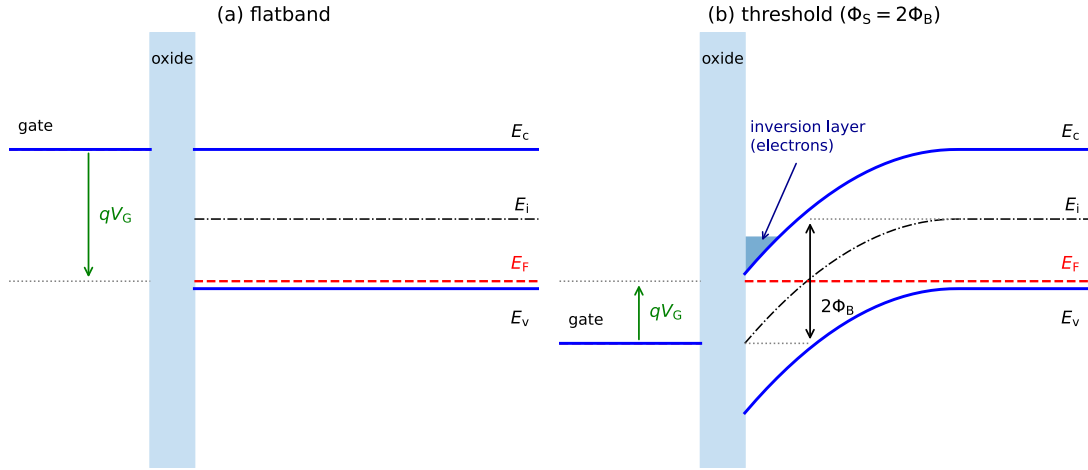


Figure 6: Simplified energy-band diagram of the MOS structure (n^{++} -poly gate, oxide, p -substrate): (a) at the flatband condition, where the bands in the silicon are flat, and (b) at the threshold condition, where the bands at the surface are bent down by $2\Phi_B$ and inversion sets in (band bending shown qualitatively). The green arrow marks the applied gate voltage V_G , which shifts the gate Fermi level by $-qV_G$ against the bulk Fermi level; in (a) $V_G = V_{FB} < 0$, in (b) $V_G = V_{th} > 0$. The blue shading in (b) marks the inversion layer of electrons at the surface.

Note 1: Example Calculation of the Threshold Voltage

Let us assume a substrate doping level of $N_a = 3 \times 10^{18} \text{ cm}^{-3}$. The Fermi potential can be calculated with Equation 8 (using the intrinsic carrier concentration of silicon $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$ at room temperature $T = 300 \text{ K}$):

$$\Phi_B = \frac{kT}{q} \ln \frac{N_a}{n_i} = 0.50 \text{ V}$$

We further use $C'_{ox} = \epsilon_{ox}/t_{ox} = 17.3 \text{ fF}/\mu\text{m}^2$ for a $t_{ox} = 2 \text{ nm}$ gate oxide. The flatband voltage for an n^+ -polysilicon gate on the p -substrate with doping N_a is (with the silicon band-gap voltage $\Phi_{bg} = 1.12 \text{ V}$)

$$\Phi_{fb} = -\frac{\Phi_{bg}}{2} - \Phi_B = -1.06 \text{ V}$$

The resulting threshold voltage is thus (*beware of units!*)

$$V_{th} = \Phi_{fb} + 2\Phi_B + \frac{\sqrt{2qN_a\epsilon_{si} \cdot 2\Phi_B}}{C'_{ox}} = 0.53 \text{ V}$$

Using these values, we can also calculate the depletion width W_{dep} in inversion, and the corresponding C'_{dep} (compare W_{dep} to t_{ox} , and C'_{dep} to C'_{ox}):

$$W_{dep} = \sqrt{\frac{2\epsilon_{si} \cdot 2\Phi_B}{qN_a}} = 21.0 \text{ nm} \quad \text{and} \quad C'_{dep} = \frac{\epsilon_{si}}{W_{dep}} = 5.0 \text{ fF}/\mu\text{m}^2$$

Using Equation 6 and Equation 2 we can now visualize the depletion width W_{dep} and the surface potential Φ_S as a function of the gate potential V_G , shown in Figure 7. Note how the surface potential saturates at $2\Phi_B$ once inversion is reached: any additional gate charge is then mirrored by electrons in the inversion layer rather than by a deeper depletion region.

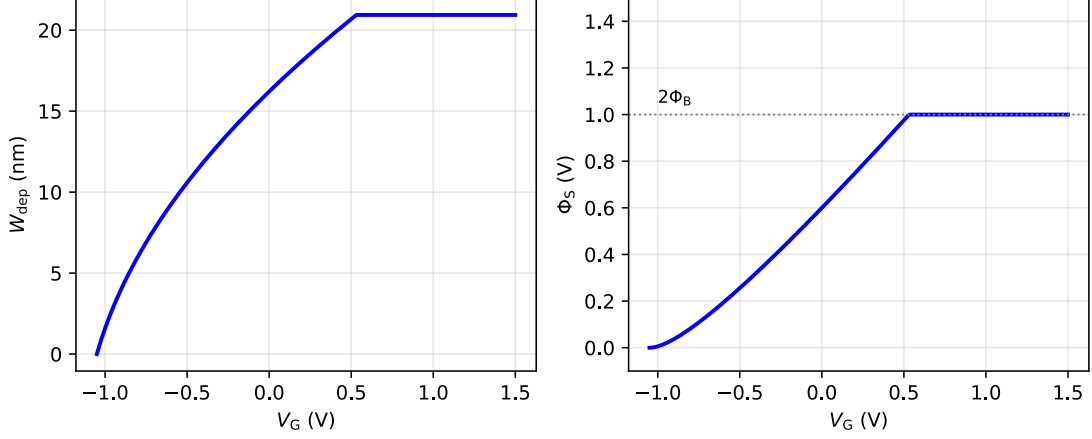


Figure 7: Depletion width W_{dep} (left) and surface potential Φ_S (right) as a function of the gate potential V_G , calculated with the depletion approximation for $N_a = 3 \times 10^{18} \text{ cm}^{-3}$, $t_{\text{ox}} = 2 \text{ nm}$, and $\Phi_{\text{fb}} = -1.05 \text{ V}$. Once inversion is reached at $\Phi_S = 2\Phi_B$ (at $V_G = V_{\text{th}} \approx 0.5 \text{ V}$), both quantities saturate.

2.2.2 Derivation of the I/V Characteristic

If we now apply a small V_{DS} , then a sufficiently large $V_{\text{GS}} > V_{\text{th}}$ builds up the inversion charge $Q'_{\text{inv}} = C'_{\text{ox}}(V_{\text{GS}} - V_{\text{th}})$ in the channel below the gate, which is then moved from source to drain by the electric field caused by V_{DS} , resulting in the drift current I_D (μ_n is the electron mobility):

$$I_D = WQ'_{\text{inv}}v = WQ'_{\text{inv}}\mu_n E = WQ'_{\text{inv}}\mu_n \frac{V_{\text{DS}}}{L} = \mu_n C'_{\text{ox}} \frac{W}{L} (V_{\text{GS}} - V_{\text{th}}) V_{\text{DS}} \quad (9)$$

If V_{DS} increases further, the channel charge $Q'_{\text{inv}}(x)$ becomes a function of the position in the channel, because the drain current causes a potential gradient dV_{cs}/dx along the channel (we assume that V_{DS} is small enough so that the whole channel is still inverted). $x = 0$ is the source end of the channel, and $x = L$ is the drain end.

$$I_D = WQ'_{\text{inv}}(x)v = WC'_{\text{ox}}(V_{\text{GS}} - V_{\text{cs}} - V_{\text{th}})\mu_n \frac{dV_{\text{cs}}}{dx}$$

Integrating along the channel from source to drain,

$$\int_0^L I_D dx = \mu_n C'_{\text{ox}} W \int_0^{V_{\text{DS}}} (V_{\text{GS}} - V_{\text{cs}} - V_{\text{th}}) dV_{\text{cs}},$$

yields the well-known **square-law** characteristic

$$I_D = \mu_n C'_{\text{ox}} \frac{W}{L} \left[(V_{\text{GS}} - V_{\text{th}}) V_{\text{DS}} - \frac{1}{2} V_{\text{DS}}^2 \right]. \quad (10)$$

Note that this simple derivation of the MOSFET I/V characteristic is only valid for long-channel devices ($L > 1 \mu\text{m}$) and does not account for second-order effects such as channel-length modulation, body effect, velocity saturation, mobility degradation with the vertical field, and subthreshold conduction. These effects will be discussed in the next section.

2.2.3 Triode Region

Equation 10 is valid only for classical, long-channel ($L > 1 \mu\text{m}$) devices, and only as long as $V_{DS} \leq V_{GS} - V_{th} = V_{od}$; this range of operation is the so-called **triode region**. The **overdrive voltage**

$$V_{od} = V_{GS} - V_{th} \quad (11)$$

is a frequently used quantity. The ratio W/L is a dimensionless quantity called the **aspect ratio** of the MOSFET—note that only the relative sizing of W and L impacts the transistor's I/V characteristic!

For $V_{DS} \ll V_{GS} - V_{th}$, we can simplify Equation 10 and see that the MOSFET behaves like a voltage-controlled resistor:

$$I_D \approx \mu_n C'_{ox} \frac{W}{L} (V_{GS} - V_{th}) V_{DS} \rightarrow g_{ds} = \frac{1}{r_o} = \frac{\partial I_D}{\partial V_{DS}} = \mu_n C'_{ox} \frac{W}{L} (V_{GS} - V_{th}) \quad (12)$$

Figure 8 shows the resulting output characteristic in the triode region for three different gate voltages.

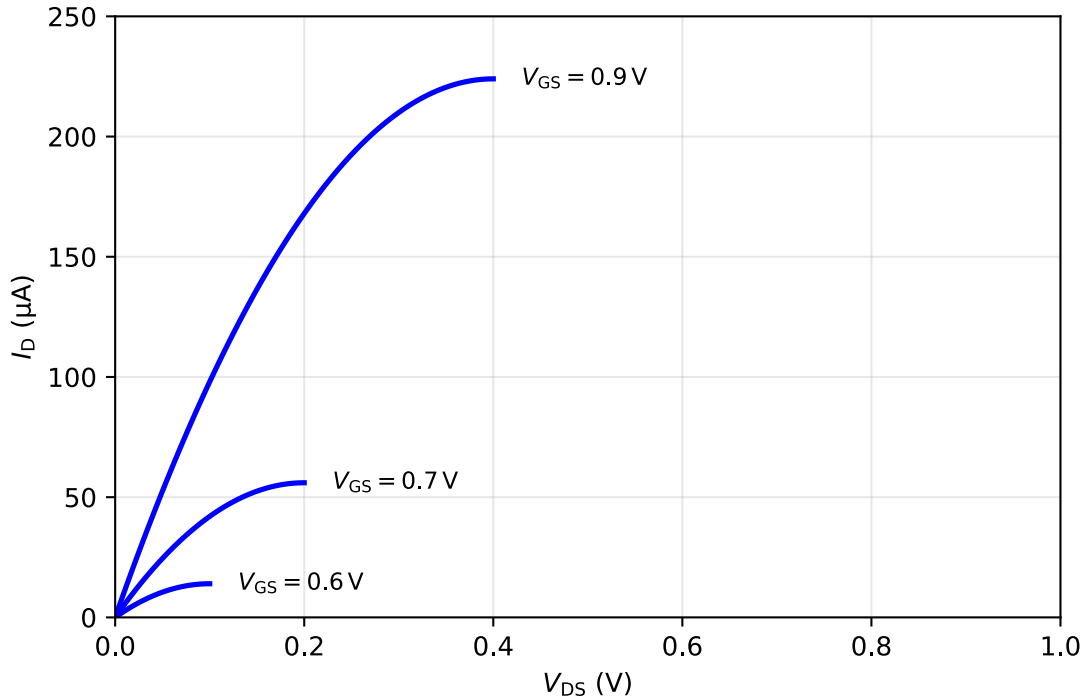


Figure 8: NMOS output characteristic in the triode region according to the square-law model ($\mu_n C'_{ox} = 280 \mu\text{A}/\text{V}^2$, $W/L = 10$, $V_{th} = 0.5 \text{V}$). Each curve ends at the edge of saturation, $V_{DS} = V_{GS} - V_{th}$.

2.2.4 Saturation Region

If $V_{DS} > V_{GS} - V_{th}$, the assumption of a completely inverted channel leading up to Equation 10 is no longer valid. The channel gets **pinched off** at the drain side; this happens at $V_{DS,sat} = V_{GS} - V_{th}$, leading to a maximum (saturated) value of the drain current. Substituting $V_{DS} = V_{DS,sat}$ into Equation 10 gives the drain current in the **saturation region**:

$$I_D = I_{DS,sat} = \frac{1}{2} \mu_n C'_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 \quad (13)$$

In the saturation region, the MOSFET behaves like a voltage-controlled current source, so it is useful to define the transconductance

$$g_m = \frac{\partial I_D}{\partial V_{GS}},$$

which can be expressed in various forms:

$$g_m = \mu_n C'_{ox} \frac{W}{L} (V_{GS} - V_{th}) = \sqrt{2 \mu_n C'_{ox} \frac{W}{L} I_D} = \frac{2 I_D}{V_{GS} - V_{th}} \quad (14)$$

We see that g_m is a linear function of the overdrive $V_{od} = V_{GS} - V_{th}$ (leading to a quadratic rise of the drain current), or a square-root function of I_D , or simply the drain current divided by half the overdrive voltage. Note that for a BJT $g_m = I_C/V_T$, so the MOSFET offers more degrees of freedom. Figure 9 shows the complete output characteristic including the boundary between triode and saturation.

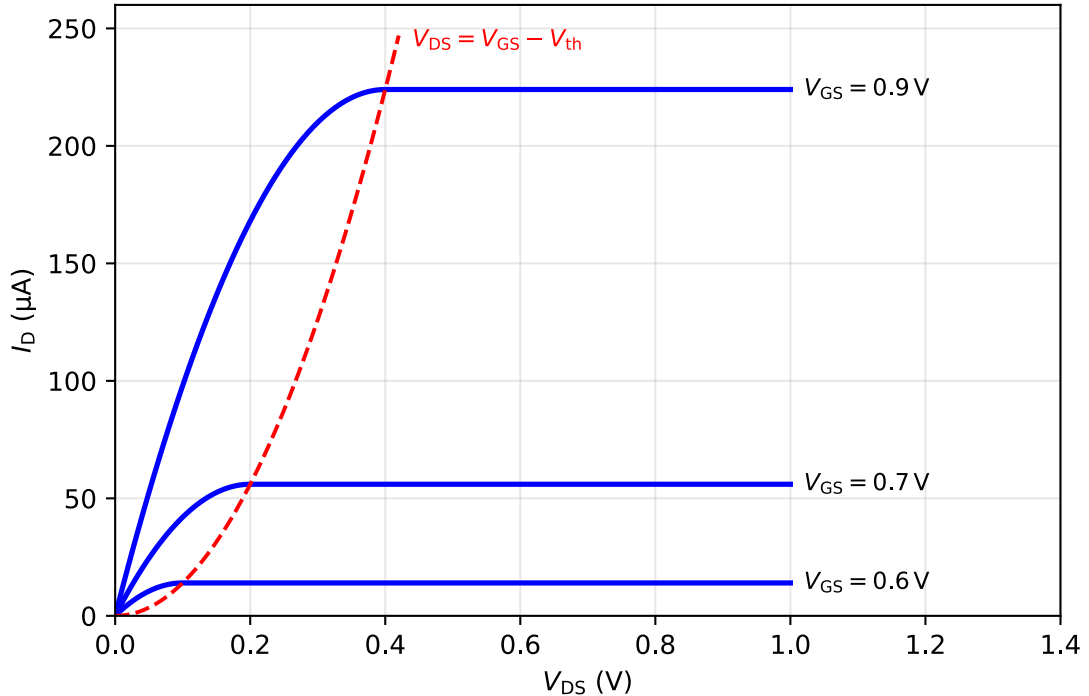


Figure 9: Complete NMOS output characteristic according to the square-law model ($\mu_n C'_{ox} = 280 \mu A/V^2$, $W/L = 10$, $V_{th} = 0.5 V$). The dashed red line marks the boundary $V_{DS} = V_{GS} - V_{th}$ between the triode and the saturation region.

2.2.5 Complementary MOS (CMOS)

In CMOS technologies, both NMOS and PMOS devices are needed and fabricated on the same wafer, as shown in Figure 10. Usually, the PMOS is located in a local n -well, and this n -well is tied to the most positive supply voltage (V_{DD}), while the p -substrate is tied to ground (V_{SS}). The substrate contacts are realized with p^+ -diffusions in the p -substrate (forming an ohmic contact), and the well contacts are realized with n^+ -diffusions in the n -well.

For increased isolation (especially for analog applications), a triple-well (or “deep n -well,” DNW) option is sometimes available; the NMOS is then located in a local p -well which is fully enclosed by an n -well and thus isolated from the common substrate.

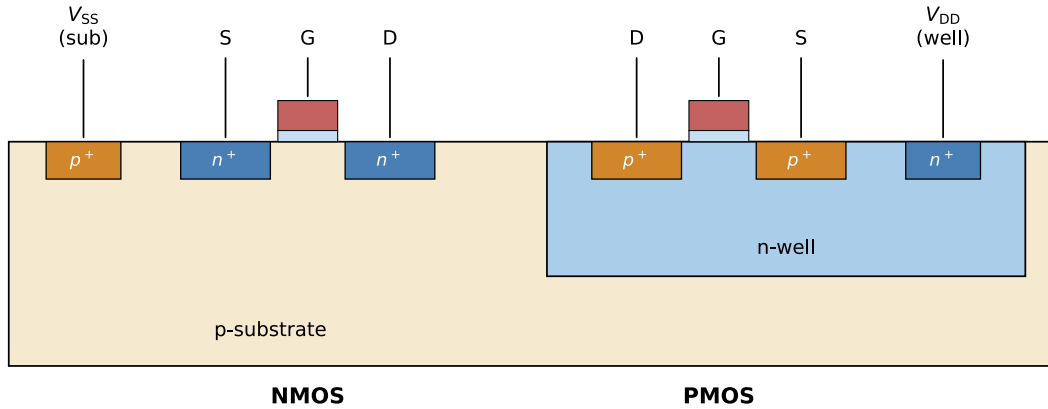


Figure 10: NMOS and PMOS transistors integrated on a single wafer: the NMOS sits directly in the p -substrate, the PMOS in a local n -well. The p^+ substrate contact ties the substrate to V_{SS} , the n^+ well contact ties the n -well to V_{DD} .

In Figure 11, we see a simple CMOS inverter consisting of a PMOS and an NMOS transistor. The PMOS is connected to the positive supply voltage V_{DD} , while the NMOS is connected to ground V_{SS} . The gates of both transistors are tied together and form the input A of the inverter, while the drains are also tied together and form the output Y . When the input is low, the PMOS is on and the NMOS is off, pulling the output high. Conversely, when the input is high, the PMOS is off and the NMOS is on, pulling the output low.

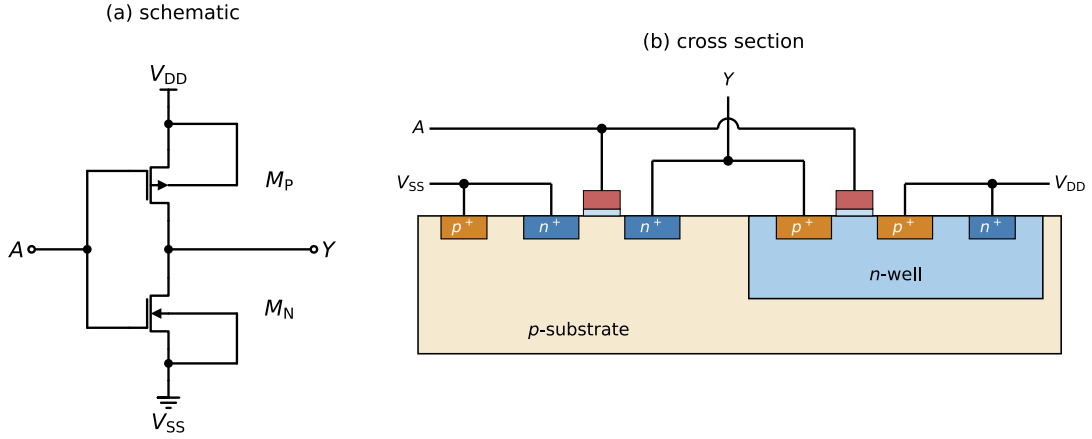


Figure 11: The CMOS inverter: (a) schematic and (b) the same circuit in the cross-section of the wafer. The PMOS pulls the output Y up when the input A is low, the NMOS pulls it down when the input is high, so in the static state no dc current flows from V_{DD} to V_{SS} . The bulk of the PMOS (its n -well) is tied to V_{DD} , the bulk of the NMOS (the p -substrate) to V_{SS} . In (b) the hop marks a wire crossing without contact, which in a real chip is realized on two different metal layers.

Note that in the CMOS technology shown in Figure 11, the bulk nodes of all the NMOS transistors are tied to the common substrate (V_{SS}), while the bulk nodes of all the PMOS transistors are tied to the n -well (V_{DD}). Since the n -wells are separated from the substrate (and can be separated between different PMOS devices), the bulk nodes of selected PMOS can be tied to different potentials than V_{DD} (e.g., to the source of the PMOS), which is called **body biasing**. This can be used to adjust the threshold voltage of the PMOS transistors or to avoid the bulk effect (see Section 2.3.1).

2.3 Second-Order Effects in the MOSFET

The simple MOSFET model developed so far is only valid to first order for long-channel devices ($L_{\min} > 1 \mu\text{m}$) under idealistic conditions. In the following, we discuss the most important second-order effects: the body effect, channel-length modulation, velocity saturation, mobility degradation with the vertical field, and subthreshold conduction.

2.3.1 Body Effect

In the derivation of Equation 7 we tied source, drain, and bulk together. Note that the inversion charge is a function of the bulk (substrate) potential V_B versus the gate potential V_G , so when the source potential V_S differs from V_B , the apparent V_{th} becomes a function of V_{SB} , because inversion is reached at a surface potential of $2\Phi_B + V_{SB}$ instead of $2\Phi_B$.

This is the so-called **body effect** (or “back-gate effect”). The effective threshold voltage is given by (V_{th0} is the threshold voltage with $V_{SB} = 0$)

$$V_{th} = V_{th0} + \gamma \left(\sqrt{2\Phi_B + V_{SB}} - \sqrt{2\Phi_B} \right) \quad (15)$$

with the body-effect coefficient γ (the factor n will be introduced in Section 2.3.5)

$$\gamma = \frac{\sqrt{2qN_a\epsilon_{si}}}{C'_{ox}} = 2(n-1)\sqrt{2\Phi_B}. \quad (16)$$

Note that for $n = 1$ (no body effect), $\gamma = 0$ and thus $V_{th} = V_{th0}$.

i Note 2: Example Calculation of the Body Effect (NMOS)

Starting with the previously used values ($N_a = 3 \times 10^{18} \text{ cm}^{-3}$, $C'_{ox} = 17.3 \text{ fF}/\mu\text{m}^2$, $\Phi_B = 0.50 \text{ V}$), we get

$$\gamma = \frac{\sqrt{2qN_a\epsilon_{si}}}{C'_{ox}} = 0.58 \text{ V}^{1/2}$$

Let us consider the originally calculated $V_{th0} = 0.53 \text{ V}$ and assume the bulk is tied to $V_{SS} = V_B = 0 \text{ V}$ while the source sits at $V_S = 0.5 \text{ V}$ (so $V_{SB} = 0.5 \text{ V}$). Using Equation 15:

$$V_{th} = 0.53 + 0.58 \left(\sqrt{2 \cdot 0.50 + 0.5} - \sqrt{2 \cdot 0.50} \right) = 0.66 \text{ V}$$

In this case, the threshold voltage has been **increased by 130 mV**. Conversely, by tying the bulk to a higher potential than the source, the threshold voltage can also be **decreased**, but beware of the risk of forward-biasing the source-bulk pn-junction (see Figure 10)!

2.3.2 Channel-Length Modulation

In the derivation of the saturation current (Equation 13), we assumed that the channel is pinched off exactly at the drain when $V_{DS} = V_{GS} - V_{th}$. When V_{DS} is increased further, this pinch-off point moves away from the drain towards the source. This can be viewed as an effective **channel-length reduction**, $L_{eff} = f(V_{DS}) < L$, and we have to use L_{eff} instead of L in Equation 13. Avoiding a direct calculation of L_{eff} , it is common to set $1/L_{eff} \approx (1 + \Delta L/L)/L$ with $\Delta L/L = \lambda V_{DS}$, so that

$$I_D = \frac{1}{2} \mu_n C'_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 (1 + \lambda V_{DS}) \quad (17)$$

This so-called **channel-length modulation** leads to a reduced output impedance of the MOSFET, i.e., a nonzero output conductance (ideally, $\lambda = 0$ and thus $g_{ds} = 0$):

$$g_{ds} = \frac{\partial I_D}{\partial V_{DS}} = I_D|_{\lambda=0} \cdot \lambda \quad (18)$$

Since $\lambda \propto 1/L$, the output resistance can be improved by increasing L beyond the minimum length, which is usually done when using the MOSFET as a current source. By scaling W along with L , the aspect ratio can be kept constant. Figure 12 shows the output characteristic including channel-length modulation; note the finite slope of the curves in saturation.

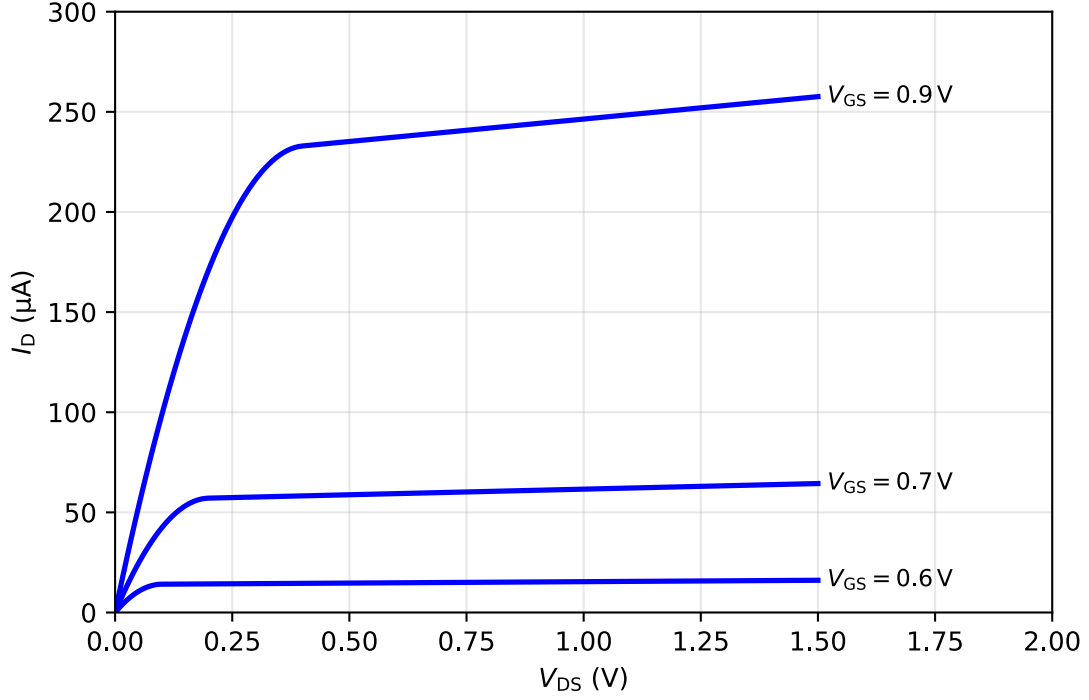


Figure 12: NMOS output characteristic including channel-length modulation ($\mu_n C'_{\text{ox}} = 280 \mu\text{A}/\text{V}^2$, $W/L = 10$, $V_{\text{th}} = 0.5 \text{ V}$, $\lambda = 0.1 \text{ V}^{-1}$). The curves in saturation now exhibit a finite slope, corresponding to a nonzero output conductance.

2.3.3 Velocity Saturation

In the derivation of Equation 9, we assumed that the velocity of the carriers in the channel is given by $v = \mu_n E_{\text{DS}}$. This is only true up to a certain speed limit of the carriers in silicon, which is around $v_{\text{sat}} = 10^7 \text{ cm/s}$. To take this **velocity saturation** into account, the following empirical model for the effective velocity can be used in the derivation:

$$v_{\text{eff}} = \begin{cases} \frac{\mu_n E_{\text{DS}}}{1 + E_{\text{DS}}/E_{\text{sat}}} & \text{if } E_{\text{DS}} \leq E_{\text{sat}} = 2v_{\text{sat}}/\mu_n \\ v_{\text{sat}} & \text{otherwise} \end{cases} \quad (19)$$

If the MOSFET enters the velocity-saturation regime, the square-law relation no longer holds, and the drain current is given by

$$I_{\text{D}} = v_{\text{sat}} C'_{\text{ox}} W (V_{\text{GS}} - V_{\text{th}}). \quad (20)$$

In velocity saturation, I_{D} is not a quadratic but a linear function of the overdrive voltage, and the transconductance hence saturates at

$$g_{\text{m,sat}} = v_{\text{sat}} C'_{\text{ox}} W \quad (21)$$

This result means that increasing the drain current beyond the velocity-saturation point does not result in an increased g_{m} , which is usually not useful.

In sub-micron technologies, it is relatively easy to reach velocity saturation: with $\mu_n = 280 \text{ cm}^2/(\text{V} \cdot \text{s})$, the resulting critical field is $E_{\text{sat}} = 7.1 \text{ V}/\mu\text{m}$, which is reached in a $0.13 \mu\text{m}$ CMOS technology at an overdrive of $V_{\text{od}} \approx 0.9 \text{ V}$ —achievable with a supply voltage of

$V_{DD} = 1.5 \text{ V}$ and $V_{th} = 0.5 \text{ V}$! Note that the stated μ_n is considerably lower than the mobility of pure silicon, because doping reduces mobility—yet another trade-off.¹

2.3.4 Mobility Degradation with Vertical Field

With high V_{GS} , the channel is confined to a narrow region at the surface, leading to more carrier scattering and thus lower mobility (remember that at the SiO_2 -silicon interface, scattering is more pronounced). We can model this effect empirically by setting

$$\mu_{n,\text{eff}} = \frac{\mu_n}{1 + \Theta(V_{GS} - V_{th})} \quad (22)$$

and using this $\mu_{n,\text{eff}}$ instead of μ_n in the saturation current (Equation 13):

$$I_D = \frac{1}{2} \frac{\mu_n C'_{ox}}{1 + \Theta V_{od}} \frac{W}{L} V_{od}^2 \approx \frac{1}{2} \mu_n C'_{ox} \frac{W}{L} (V_{od}^2 - \Theta V_{od}^3) \quad (23)$$

This introduces a *cubic* nonlinearity into an otherwise quadratic relationship, which can be relevant for precision analog and high-frequency circuits, with $\Theta \approx (10^{-9} \text{ m/V})/t_{ox} = 0.5 \text{ V}^{-1}$ for $t_{ox} = 2 \text{ nm}$.

2.3.5 Subthreshold Conduction

In the derivation of Equation 7, we assumed that inversion sets in *abruptly* when the surface potential reaches $2\Phi_B$. However, even before reaching this level, there are already electrons present in the channel (for the NMOS; holes for the PMOS), which lead to conduction between drain and source even for $V_{GS} < V_{th}$.

This mode of operation is called **subthreshold conduction**, and it leads to a leakage current based on diffusion when the MOSFET should be off. The drain current in this regime is described by

$$I_D = I_0 \frac{W}{L} e^{\frac{V_{GS}}{nV_T}} \left(1 - e^{-\frac{V_{DS}}{V_T}}\right) = \mu_n C'_{ox} \frac{W}{L} 2nV_T^2 e^{\frac{V_{GS}-V_{th}}{nV_T}} \left(1 - e^{-\frac{V_{DS}}{V_T}}\right). \quad (24)$$

The nonideality factor (or slope factor) n is defined by (see Equation 16)

$$n = 1 + \frac{C'_{dep}}{C'_{ox}} \quad (25)$$

and is on the order of 1 (perfect MOSFET) to 2 (pretty bad). Using the values of C'_{ox} and C'_{dep} from the previous examples, we get $n = 1.29$. $V_T = kT/q$ is the well-known thermal voltage of 25.8 mV at 300 K, and

$$I_0 = \mu_n C'_{ox} 2nV_T^2 e^{-\frac{V_{th}}{nV_T}}.$$

Note that the depletion capacitance C'_{dep} is given by Equation 3 evaluated at $\Phi_S = 2\Phi_B$:

¹The mobility value used here is not fully consistent with the process constant $K_n = \mu_n C'_{ox} = 280 \mu\text{A/V}^2$ in Table 1, which (with $C'_{ox} = 17.3 \text{ fF}/\mu\text{m}^2$) would imply an effective $\mu_n \approx 160 \text{ cm}^2/(\text{V} \cdot \text{s})$. The effective mobility depends strongly on the vertical field and thus on bias; such spread between simple hand-calculation parameters is normal and is one more reason to verify all sizing choices by simulation.

$$C'_{\text{dep}} = \sqrt{\frac{qN_a\epsilon_{\text{si}}}{2 \cdot 2\Phi_B}}$$

The inversion condition used in the derivation of the threshold voltage in Equation 7 is the point where *the electron concentration at the surface equals the bulk doping concentration* N_a —but even before this condition is met, there are electrons in the channel. Another way to look at this is that the surface potential effectively forward-biases the channel-source junction (since the channel region is p -type and the source is n^+), which can be seen in the plot of Φ_S in Figure 7. When the forward bias of this junction is gradually increased, electrons are injected into the channel area (similar to a BJT), moving across the channel by diffusion, and collected by the drain. At the point where the channel switches from p - to n -type due to inversion, this “diode” approaches an ohmic connection linking drain and source.

Although this **weak inversion** mode of the MOSFET is generally unwanted in digital circuits (as it leads to leakage currents), it is very useful for low-power analog (and digital) operation. Assuming $V_{\text{DS}} > 4V_T \approx 100 \text{ mV}$, then

$$I_D \approx I_0 \frac{W}{L} e^{\frac{V_{\text{GS}}}{nV_T}}$$

and the resulting transconductance is

$$g_m = \frac{\partial I_D}{\partial V_{\text{GS}}} = \frac{I_D}{nV_T} \quad (26)$$

This transconductance in weak inversion is similar to that of a BJT, except for the reduction by the factor n !

2.3.6 MOSFET I/V Characteristic Including the Body Effect

In the earlier derivation of the square law (Equation 10), we made a deliberate error: when the channel potential V_{cs} departs from V_S towards V_D , we need to take the body effect along the channel into account by using nV_{cs} instead of V_{cs} , so $Q'_{\text{inv}}(x) = C'_{\text{ox}}(V_{\text{GS}} - nV_{\text{cs}} - V_{\text{th}})$. The corrected derivation is then

$$\int_0^L I_D dx = \mu_n C'_{\text{ox}} W \int_0^{V_{\text{DS}}} (V_{\text{GS}} - nV_{\text{cs}} - V_{\text{th}}) dV_{\text{cs}}$$

resulting in

$$I_D = \mu_n C'_{\text{ox}} \frac{W}{L} \left[(V_{\text{GS}} - V_{\text{th}}) V_{\text{DS}} - \frac{n}{2} V_{\text{DS}}^2 \right] \quad (27)$$

In the deep-triode region, there is no difference from the previous result, as the term $nV_{\text{DS}}^2/2$ vanishes for small V_{DS} . The saturation voltage is now $V_{\text{DS,sat}} = (V_{\text{GS}} - V_{\text{th}})/n$, and the drain current in saturation becomes (compare to Equation 13)

$$I_D = I_{\text{DS,sat}} = \frac{1}{2n} \mu_n C'_{\text{ox}} \frac{W}{L} (V_{\text{GS}} - V_{\text{th}})^2. \quad (28)$$

The corrected expressions for the transconductance are

$$g_m = \mu_n C'_{ox} \frac{W}{L} \frac{V_{GS} - V_{th}}{n} = \sqrt{\frac{2}{n}} \mu_n C'_{ox} \frac{W}{L} I_D = \frac{2I_D}{V_{GS} - V_{th}}. \quad (29)$$

2.4 MOSFET AC Operation

So far, we have only considered the dc operation of the MOSFET, but the physical construction of the transistor leads to unwanted capacitances and resistances, which need to be accounted for in dynamic operation. The most important capacitive elements are:

- The oxide capacitance between the gate and the channel, $C_{gg} = C'_{ox} WL$.
- The gate-drain and gate-source overlap capacitances $C_{ov} = C'_{ov} W$.
- The junction capacitance between drain/source and substrate, $C_j = C_{j0}/(1 + V_{DB|SB}/\Phi_B)^m$ with $m \approx 0.3...0.4$. C_{j0} is defined by the area and perimeter of the drain/source regions.
- The depletion capacitance between the channel (the inversion layer) and the substrate, $C_{dep} = C'_{dep} WL$.

To identify the dominant capacitive elements between gate-source (C_{GS}) and gate-drain (C_{GD}), we need to distinguish between the triode and the saturation region, see Table 2.

Table 2: Dominant gate capacitances of the MOSFET in the triode and saturation regions

Operating region	C_{GS}	C_{GD}
Triode	$\frac{1}{2}C_{gg} + C_{ov}$	$\frac{1}{2}C_{gg} + C_{ov}$
Saturation	$\frac{2}{3}C_{gg} + C_{ov}$	C_{ov}

In addition to these capacitances, the resistance of the gate connection is usually dominant among the parasitic resistances. The end-to-end resistance of the gate stripe is $R_s \cdot W/L$ (with the sheet resistance R_s set by the gate material). Due to the distributed nature of the gate along the width of the transistor, a reasonable lumped model for a single-sided gate contact is

$$R_G = \frac{1}{3}R_s \frac{W}{L} \quad (30)$$

Using a **double-sided gate contact**, the effective gate resistance is reduced to

$$R_G = \left(\frac{1}{3}R_s \frac{W}{2 \cdot L} \right) \parallel \left(\frac{1}{3}R_s \frac{W}{2 \cdot L} \right) = \frac{1}{12}R_s \frac{W}{L} \quad (31)$$

The impact of R_G on performance can be further minimized by constructing a large transistor out of m smaller transistors connected in parallel (“fingers”). If the total width is $W = mW'$ (with each finger having width W'), the gates of all unit transistors are connected in parallel, and the effective single-sided gate resistance becomes

$$R_{G,eff} = \frac{R'_G}{m} = \frac{\frac{1}{3}R_s \frac{W}{m \cdot L}}{m} = \frac{1}{m^2}R_G \quad (32)$$

A typical analog/RF layout therefore splits a wide transistor into several fingers of limited length (a few μm), uses double-sided gate contacts, and surrounds the device with a closed well-contact ring; the multi-finger layout technique is discussed in detail in **@sec-layout-analog**.

2.5 MOSFET Small-Signal Model

The complete small-signal model of a MOSFET in saturation is shown in Figure 13. The transconductance g_m is modeled by a voltage-controlled current source, as is the body effect, which contributes a back-gate transconductance

$$g_{mb} = g_m \frac{\gamma}{2\sqrt{2\Phi_B + V_{SB}}}.$$

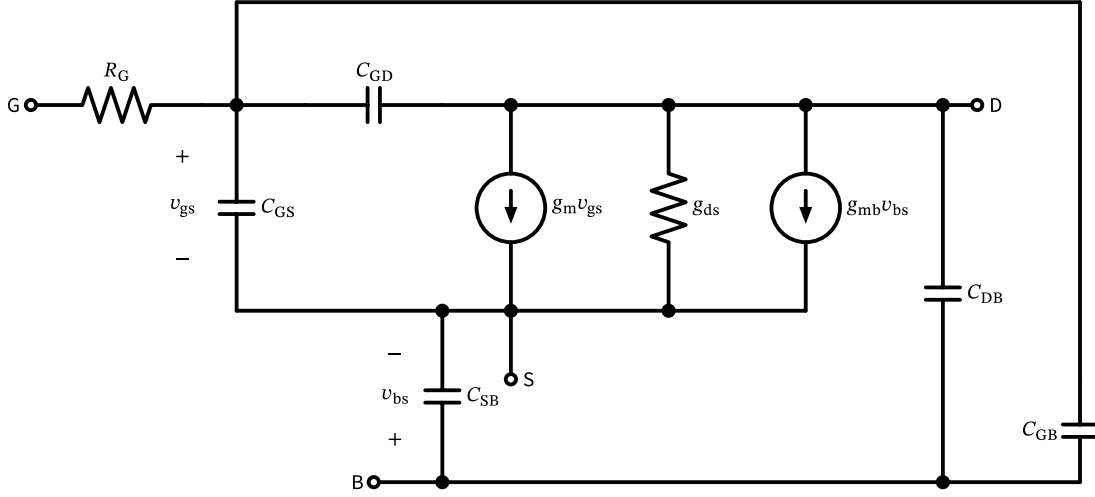


Figure 13: Complete small-signal model of the MOSFET in saturation, including the gate resistance, the capacitances to all terminals, the transconductance g_m , the back-gate transconductance g_{mb} , and the output conductance g_{ds} .

2.5.1 High-Frequency Operation

For high-frequency operation, the definition of the transit frequency f_T is useful; this is the frequency where the magnitude of the current gain $|I_{ds}|/|I_{gs}|$ falls to one. The ac drain current can be calculated based on the small-signal model:

$$I_{ds}(j\omega) = I_{gs}(j\omega) \frac{g_m - j\omega C_{GD}}{j\omega(C_{GS} + C_{GD})}$$

The transit frequency is thus (extrapolating the low-frequency transfer function)

$$\omega_T = 2\pi f_T \approx \frac{g_m}{C_{GS} + C_{GD}} \approx \frac{g_m}{C_{GG}} \approx \frac{3}{2n} \frac{\mu}{L^2} (V_{GS} - V_{th}). \quad (33)$$

In addition, the maximum oscillation frequency f_{max} is a useful metric; it is the frequency where the maximum available power gain drops to one (MAG = 0 dB):

$$f_{max} \approx \sqrt{\frac{f_T}{8\pi R_G C_{GD}}} \quad (34)$$

2.5.2 Simplified Small-Signal Models

In most cases, a simplified model is sufficient for hand calculations. Neglecting most model elements, the two models shown in Figure 14 can be used in the saturation region. The left π -model is usually better suited for circuits with the signal input at the gate (like in a common-

source or common-drain amplifier), whereas the right τ -model is advantageous when the input is at the source (like in a common-gate amplifier).

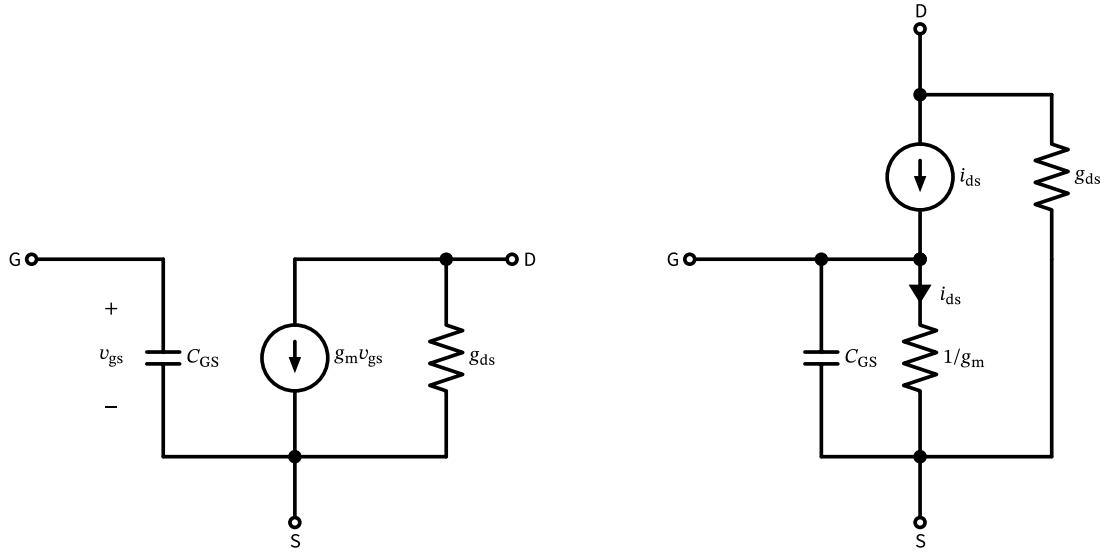


Figure 14: Simplified small-signal models of the MOSFET in saturation for hand calculations: π -model (left), preferred when the input is at the gate, and τ -model (right), preferred when the input is at the source. In the τ -model, the current through the resistor $1/g_m$ equals i_{ds} , which is mirrored by the controlled source into the drain.

2.6 MOSFET Scaling

Making MOSFETs smaller has benefits for power, performance, and area (PPA). Traditionally, every 2–3 years a scaled (= smaller) CMOS technology node has been introduced. In Figure 15, the technology nodes are plotted versus their year of first volume production; note the logarithmic axis!

In the beginning, the node name stated the minimum feature size of the given technology, which is usually the minimum gate length ($L_{\min}$): in a 130 nm technology, $L_{\min} = 130$ nm. This holds well down to about the 90 nm node; from there on the printed gate length is already shorter than the node name (the 65 nm node used a physical gate length of roughly 35 nm), and from the 22/20 nm node onwards the technology name is merely a “label.” Since traditionally a scaling factor of $\sqrt{2}$ has been used, the node names reflect this: 350 $\rightarrow$ 250 $\rightarrow$ 180 $\rightarrow$ 130 $\rightarrow$ 90 $\rightarrow$ 65 $\rightarrow$ 45 $\rightarrow$ 32/28 $\rightarrow$ 22/20 $\rightarrow$ 16/14 $\rightarrow$ 10 $\rightarrow$ 7 $\rightarrow$ 5 $\rightarrow$ 3 $\rightarrow$ 2 $\rightarrow$ 1.6 nm.

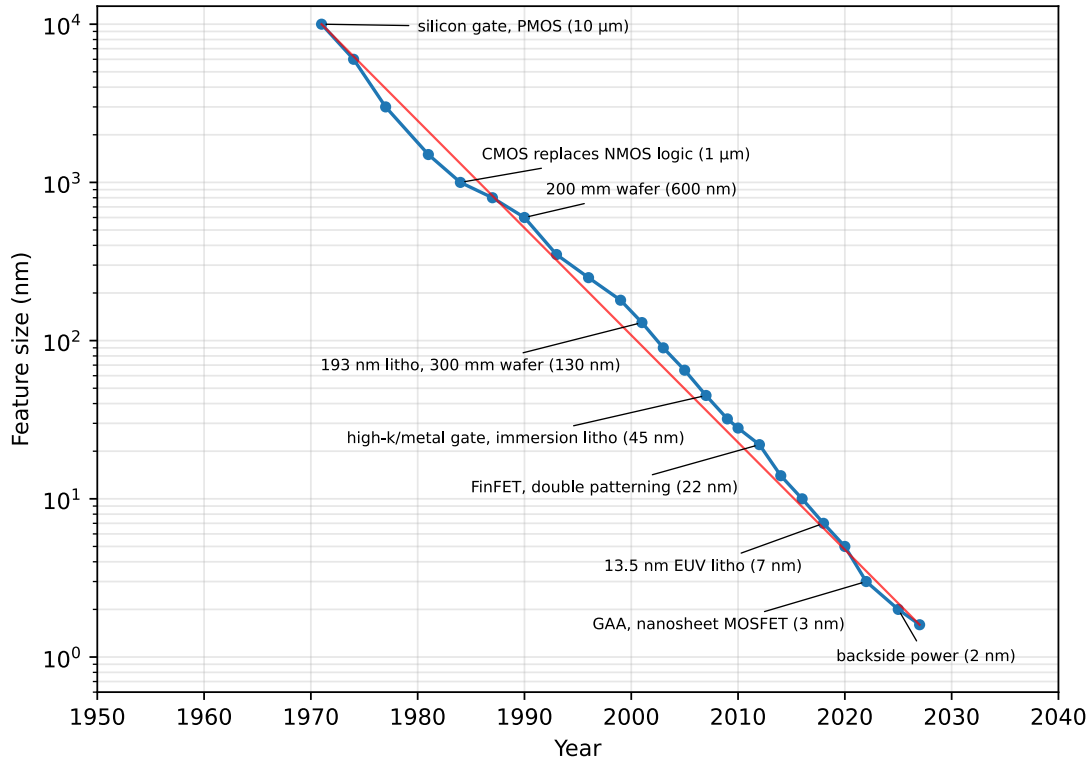


Figure 15: CMOS technology scaling trend: minimum feature size (node name) versus year of first volume production, with key technology innovations annotated (data compiled from public sources, e.g., wikichip.org).

2.6.1 Scaling Theory

The ideal scaling theory follows three rules, formulated by R. Dennard in 1974 [7], to generate the next CMOS technology node from the current one:

1. Reduce all lateral and vertical dimensions by α ($\alpha = \sqrt{2}$ for classical scaling; the area of a MOSFET thus *shrinks* by $\alpha^2 = 2$).
2. Reduce the threshold voltage V_{th} and the supply voltage V_{DD} by α .
3. Increase all doping levels by α .

This scaling is called **constant-field scaling**. If we apply these rules, the MOSFET parameters in the shrunk technology (marked by *) scale as follows. The drain saturation current is reduced by α :

$$I_D^* = \frac{1}{2} \mu_n (\alpha C'_{ox}) \frac{W/\alpha}{L/\alpha} \left(\frac{V_{GS}}{\alpha} - \frac{V_{th}}{\alpha} \right)^2 = \frac{1}{\alpha} I_D$$

This is offset by the fact that the input capacitance is also reduced:

$$C_{gg}^* = \frac{W}{\alpha} \frac{L}{\alpha} (\alpha C'_{ox}) = \frac{1}{\alpha} C_{gg}$$

To estimate the scaling of the gate delay of logic gates built from scaled devices, we use the simple model $T_d \propto C_{gg} V_{DD} / I_D$, which relates the gate delay to the time it takes to charge or discharge the load capacitance of a driven gate between the supply rails. The scaled delay is

$$T_d^* = \frac{C_{gg}/\alpha}{I_D/\alpha} \frac{V_{DD}}{\alpha} = \frac{1}{\alpha} T_d$$

The speed of digital circuits thus benefits from scaling, as the gate delay shrinks in each new technology node! Using a simple model for the dynamic power consumption of CMOS logic, $P_{\text{dyn}} = f_{\text{clk}} C_{\text{dyn}} V_{DD}^2$ (where f_{clk} is the clock frequency and $C_{\text{dyn}} \propto C_{gg}$ is the total switched capacitance), we see that

$$P_{\text{dyn}}^* = f_{\text{clk}} \frac{C_{\text{dyn}}}{\alpha} \left(\frac{V_{DD}}{\alpha} \right)^2 = \frac{1}{\alpha^3} P_{\text{dyn}}$$

so the dynamic power consumption of a scaled logic circuit benefits dramatically. Note that the power per function, clocked at the same frequency, scales by α^{-3} —but since we can also increase the clock frequency by α , and can integrate α^2 more gates per area, the total power consumption would stay constant if we fully exploited the scaling benefit. To improve power, performance, *and* area (PPA) simultaneously, a trade-off has to be found.

2.6.2 Analog Scaling

While there is a clear scaling benefit for digital circuits, the benefits for analog circuits are less clear. Generally, the reduction of V_{DD} is a drawback, since it likely reduces the dynamic range when processing voltage signals. The transconductance scales like

$$g_m^* = \mu_n (\alpha C'_{ox}) \frac{W/\alpha}{L/\alpha} \frac{V_{GS} - V_{th}}{\alpha} = g_m$$

so it seems to be scale-invariant. However, there is a benefit in device speed, as the transit frequency scales like

$$f_T^* = \frac{1}{2\pi} \frac{g_m}{C_{gg}/\alpha} = \alpha f_T$$

so the speed of the MOSFET increases, opening the door to better analog and RF performance, albeit at lower supply voltages.

2.6.3 Scaling Summary

Table 3 summarizes the constant-field scaling factors derived above. Scaling equations fitted to real technology data from 180 nm down to 7 nm can be found in [8].

Table 3: Constant-field (Dennard) scaling of MOSFET parameters

Parameter	Symbol	Scaling factor
Gate length	$L_{\min}$	α^{-1}
MOSFET area	WL	α^{-2}
Gate oxide thickness	t_{ox}	α^{-1}
Substrate doping	N_{a}	α
Drain current in saturation	I_{D}	α^{-1}
Gate capacitance	C_{GS}	α^{-1}
Supply voltage	V_{DD}	α^{-1}
Threshold voltage	V_{th}	α^{-1}
Gate delay	T_{d}	α^{-1}
Clock frequency	f_{clk}	α
Dynamic power consumption per gate	P_{dyn}	α^{-3}

2.6.4 Interconnect Scaling

As the MOSFETs get smaller, the metal wiring also needs to scale to finer dimensions. We assume that, along with the lateral dimensions, the metal thickness and the dielectric inter-metal insulation scale down as well. The resistance per length of a scaled metal wire (with thickness t and width w) scales as

$$\frac{R^*}{l} = \rho \frac{1}{(t/\alpha)(w/\alpha)} = \alpha^2 \frac{R}{l}$$

so the wires get significantly more resistive! The wire-to-wire fringe capacitance per length for a minimum-width wire with minimum spacing s scales as

$$\frac{C_{\text{f}}^*}{l} = \epsilon_{\text{ox}} \frac{t/\alpha}{s/\alpha} = \frac{C_{\text{f}}}{l}$$

and the vertical coupling capacitance per length (with insulator thickness h) scales as

$$\frac{C_{\text{p}}^*}{l} = \epsilon_{\text{ox}} \frac{w/\alpha}{h/\alpha} = \frac{C_{\text{p}}}{l}$$

so both capacitance values are, to first order, *invariant to scaling*. The RC wire delay per squared length is thus

$$\frac{T_{\text{wire}}^*}{l^2} \propto \frac{R^*}{l} \left(\frac{C_{\text{f}}^*}{l} + \frac{C_{\text{p}}^*}{l} \right) = \alpha^2 \frac{T_{\text{wire}}}{l^2}$$

For short routes, this increased relative delay is offset by the length scaling, so that

$$T_{\text{wire}}^* \propto \left(\frac{l}{\alpha}\right)^2 \frac{T_{\text{wire}}^*}{l^2} = T_{\text{wire}}$$

For long routes, however, the wire delay increases significantly. To cope with this delay, other solutions are needed: using multiple wiring levels (lower metals levels scaled for density, higher ones un-scaled for speed), or adapting the architecture, e.g., pipelining long-range data transfers.

2.7 MOSFET Short-Channel Effects

As the MOSFET gate length is scaled below $1 \mu\text{m}$, various imperfections kick in. We discuss subthreshold conduction and leakage, gate oxide leakage, and drain-induced barrier lowering (DIBL). Other effects, like random dopant fluctuation (RDF), poly depletion, or quantum-mechanical effects, are not covered here.

2.7.1 Subthreshold Leakage

Revisiting Equation 24 and neglecting the V_{DS} -dependent term, we can re-formulate the subthreshold drain current as

$$I_{\text{D}} = 100 \text{ nA} \cdot \frac{W}{L} e^{\frac{V_{\text{GS}} - V_{\text{th}}}{n V_{\text{T}}}} \quad (35)$$

Here, V_{th} has been re-defined as the V_{GS} at which $I_{\text{D}} = 100 \text{ nA} \cdot W/L$, which is a quite practical definition for leakage current discussions. The nonideality factor can also be written as

$$n = 1 + \frac{C'_{\text{dep}}}{C'_{\text{ox}}} = 1 + \underbrace{\frac{\varepsilon_{\text{si}}}{\varepsilon_{\text{ox}}} \frac{t_{\text{ox}}}{W_{\text{dep}}}}_{\approx 3} \quad (36)$$

Using Equation 35, we see that the leakage current in the off-state ($V_{\text{GS}} = 0$) is impacted by only *two* parameters (W/L is usually fixed by the circuit design):

$$I_{\text{leak}} = 100 \text{ nA} \cdot \frac{W}{L} e^{-\frac{V_{\text{th}}}{n V_{\text{T}}}} \quad (37)$$

1. The threshold voltage V_{th} : a lower V_{th} increases the overdrive when the MOSFET is turned on and thus reduces the gate delay, but it exponentially increases the leakage current.
2. The nonideality factor n (ideally, $n = 1$).

2.7.2 Subthreshold Slope

To quantify the “turn-off” behavior of a MOSFET, the **subthreshold slope** S is defined, starting from Equation 35, and consisting of the two only parameters V_{T} and n :

$$S = \left[\frac{\partial(\log_{10} I_{\text{D}})}{\partial V_{\text{GS}}} \right]^{-1} = \frac{n V_{\text{T}}}{\log_{10}(e)} = 2.3 n V_{\text{T}} \quad (38)$$

The subthreshold slope expresses which change of V_{GS} lowers (or increases) the drain current by a factor of ten. Ideally, S should be as small as possible, meaning that the MOSFET can be turned off rapidly with a small change of V_{GS} :

- The lower bound is $S = 2.3 V_{\text{T}} = 59.4 \text{ mV/dec}$ at 300 K.

- Using $n = 1.29$ from the previous example, $S = 76.6 \text{ mV/dec}$, which is a good value; S for bulk CMOS is in the range of 80 to 120 mV/dec.

The impact of the subthreshold slope essentially stopped the scaling of V_{DD} and V_{th} , as the leakage current would otherwise become impractically large. This issue is exacerbated by the temperature dependence of the threshold voltage of $\approx -1 \text{ mV/K}$ and its process variation on the order of $\pm 50 \text{ mV}$. A threshold-voltage shift of a hundred millivolts may not sound like much, but it enters the leakage exponentially and thus has a drastic impact! Figure 16 shows how the supply voltage stopped scaling at around 0.8 V.

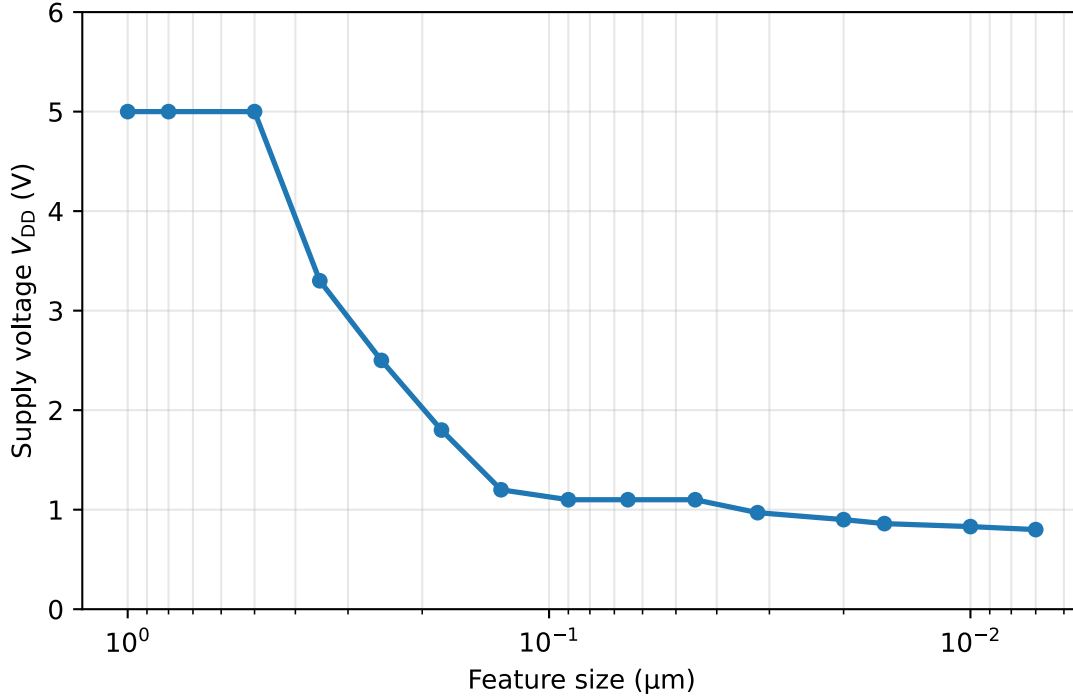


Figure 16: Nominal core supply voltage versus CMOS technology node. Below roughly 90 nm, the supply voltage no longer scales with the feature size but flattens out around 1 V, limited by the subthreshold leakage (data after Stillmaker and Baas, 2017).

2.7.3 Gate Oxide Leakage

As can be seen in Equation 36, to minimize n (and thus the subthreshold slope), and also to increase the drain current in saturation, C'_{ox} needs to be maximized. This has driven the scaling down of the gate oxide thickness, as discussed in the scaling rules. Over the years, the oxide thickness has been reduced from 300 nm to 1.2 nm, but severe issues appear once the oxide is only a few nanometers thick:

- The breakdown voltage of the oxide is reduced, mandating a low V_{DD} .
- The gate leakage current increases dramatically due to quantum-mechanical tunneling.

An alternative is needed to increase C'_{ox} without further reducing t_{ox} : the introduction of alternative gate dielectrics with increased permittivity, the so-called **high-k gate dielectrics**. One suitable material is hafnium dioxide: while SiO_2 has $\epsilon_r = 3.9$, HfO_2 offers $\epsilon_r = 24$. However, there are potential manufacturing issues, like chemical reactions with the silicon

substrate and the gate material, reliability concerns, and reduced surface mobility. Still, high- κ dielectrics are now widely used in modern CMOS processes.

2.7.4 Drain-Induced Barrier Lowering (DIBL)

In scaled MOSFETs, there exists another parasitic effect that lowers V_{th} as a function of L , called **drain-induced barrier lowering (DIBL)**. (Remember that lowering V_{th} has an exponential effect on the leakage current!) This lowering of V_{th} with *decreasing* L is also called the short-channel effect (SCE). Note that there is also a *reverse* short-channel effect (RSCE), which *increases* V_{th} with decreasing L , caused by halo implants around drain and source in modern CMOS processes. Effectively, $V_{th} = f(L)$ can be a wild function of the MOSFET length, so devices will only match if constructed with the same L ! Note that V_{th} also depends on V_{DS} due to DIBL, and on device W , as well as temperature, and other second-order effects, so ultimately $V_{th} = f(L, W, V_{DS}, T)$!

The impact of L and V_{DS} on V_{th} can be modeled by

$$\Delta V_{th} = -(V_{DS} + 0.4 \text{ V}) e^{-L/l_d} \quad (39)$$

and is illustrated in Figure 17. The DIBL characteristic length is $l_d \propto \sqrt[3]{t_{ox} W_{dep} X_j}$. To reduce l_d , one or all of these parameters must be reduced:

- The oxide thickness t_{ox} , which has already been discussed in the context of gate leakage.
- The depletion width W_{dep} .
- The drain junction depth X_j .

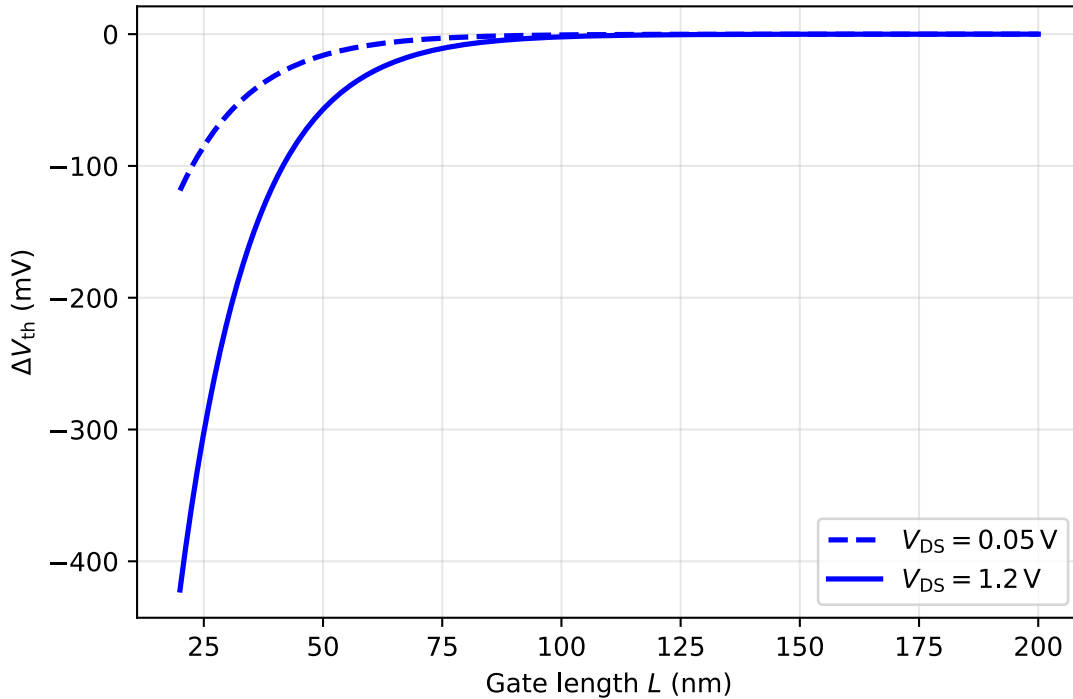


Figure 17: Threshold-voltage shift ΔV_{th} due to DIBL as a function of gate length, calculated with the model of Equation 39 for a characteristic length $l_d = 15 \text{ nm}$. For short gate lengths and high drain voltages, the threshold voltage drops rapidly (compare, e.g., the measured data for a 65 nm technology in [9]).

Reducing the depletion width and the junction depth requires increasing N_a (see Equation 2). However, increasing N_a also increases V_{th} (see Equation 7), which must be counteracted by increasing C'_{ox} —but there are limits due to gate leakage and breakdown, as discussed before. In addition, increasing N_a increases the subthreshold slope (see Equation 36) and lowers the mobility due to dopant scattering. So an alternative to ever-increasing substrate doping had to be found.

2.7.5 Solving DIBL: The Ultra-Thin-Body MOSFET

As scaling up N_a to constrain the *vertical* dimensions has reached its limits, an alternative solution is to *remove the silicon* under the channel to *physically* constrain W_{dep} and X_j . In the ultra-thin-body (UTB) MOSFET, the silicon channel is realized on top of a layer of SiO_2 , so the depletion and junction depths are limited to the silicon film thickness $t_{si} \leq L/4$. This is called **silicon-on-insulator (SOI)** technology; the structure is shown in Figure 18 (a). Additional advantages are:

- Since much less body doping N_a is needed, the carrier mobility is higher.
- There is no (or only a small) body effect, since the body is floating and fully depleted.
- The drain/source-to-bulk capacitance is largely reduced and almost **constant**.

The disadvantage is the higher cost of the SOI wafers, as manufacturing the Si- SiO_2 -Si stack is expensive.

2.7.6 Solving DIBL: FinFET and Nanosheet

Another alternative is to tilt the silicon channel by 90° , so that it sticks out of the wafer like a “fin,” and to wrap the gate around this fin on three sides: the **FinFET** (or tri-gate MOSFET), shown in Figure 18 (b). FinFETs have a few important differences compared to bulk MOSFETs:

- The device width is quantized, with $W = W_F + 2H_F$ per fin (W_F is the width and H_F the height of the fin), so the circuit design style must be adapted (the design parameter is the number of fins N_F instead of W).
- Drain and source are contacted at the ends of the fin, so the series resistances R_S and R_D are higher, degrading the transconductance due to series feedback: $g_{m,eff} = g_m / (1 + g_m R_S)$.
- Since the subthreshold slope is improved (smaller), the leakage current is lower, which allows lowering V_{th} and V_{DD} .
- DIBL and thus g_{ds} are lower, so the MOSFET self-gain g_m/g_{ds} is high.
- There is practically no body effect, and due to the low channel doping the mobility is high, leading to improved g_m/I_D and $\mu_n \approx \mu_p$.
- The silicon fins are isolated and surrounded by SiO_2 , so self-heating of the fin can be an issue.
- The 3D structure increases the coupling capacitance C_{ov} .
- A drawback is that SRAM and digital standard cells often use minimum-size transistors, and for yield reasons a minimum-sized FinFET requires two fins—a strong constraint when, e.g., the PMOS needs to be sized differently from the NMOS (the next step up is three fins, a large quantization step).

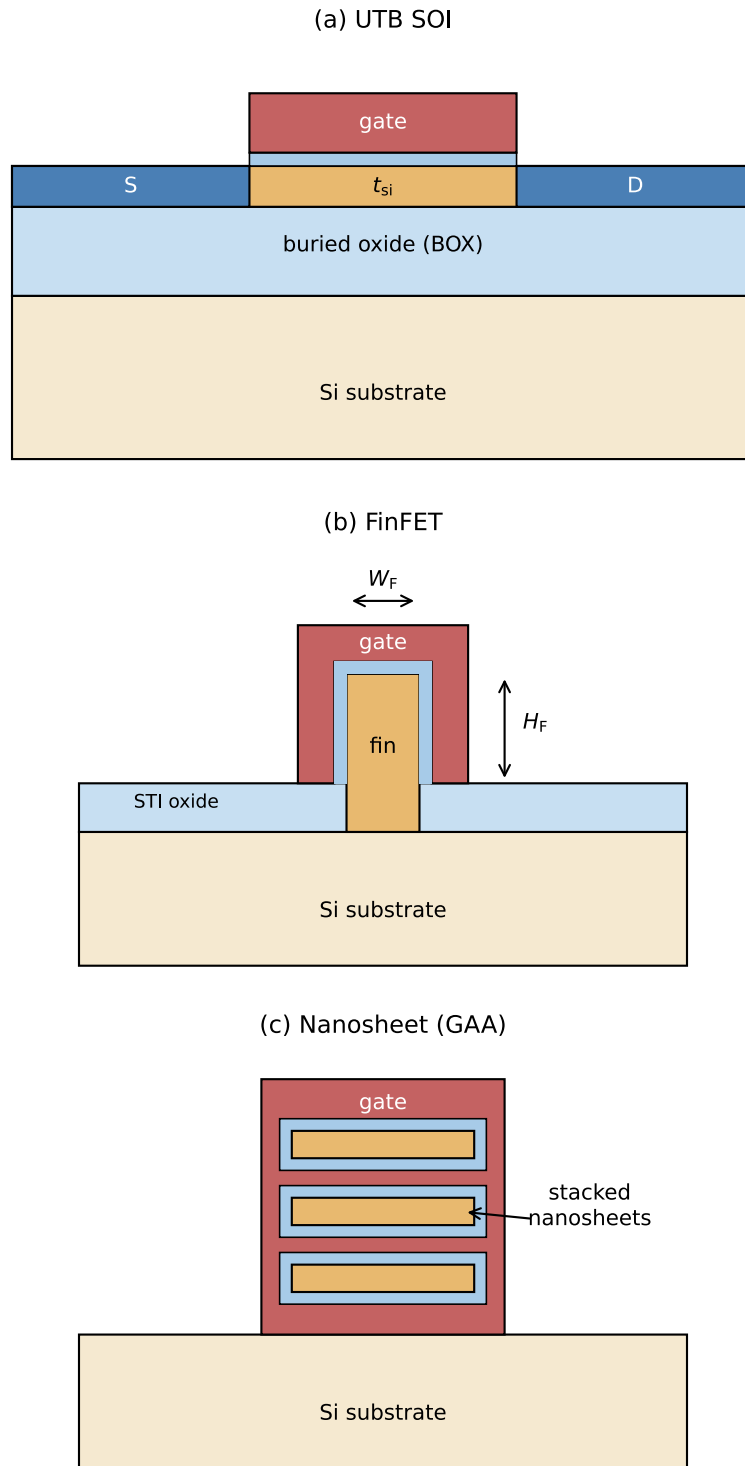


Figure 18: Advanced MOSFET structures with improved electrostatic gate control: (a) ultra-thin-body (UTB) SOI MOSFET shown along the channel, (b) FinFET shown perpendicular to the channel (the gate wraps around the fin on three sides), and (c) nanosheet (gate-all-around) MOSFET, where stacked channels are completely surrounded by the gate.

The evolution of the FinFET is the **nanosheet MOSFET**, first introduced at the 3 nm node (and used by all foundries at 2 nm) and shown in Figure 18 (c). One advantage is the even

better electrostatic control of the gate over the channel, as the stacked channels are completely wrapped by the gate—hence also called gate-all-around (GAA) MOSFET. In addition, the stacked channels can be either *n*- or *p*-type (or mixed), and could even use different materials, like SiGe or GaAs, opening the door to 3D integration.

2.8 Simulated Characteristics in IHP SG13

To connect the simple models we derived earlier to reality, Figure 19 to Figure 22 show simulated I/V characteristics of NMOS and PMOS devices ($W = 1.3 \mu\text{m}$, $L = 0.13 \mu\text{m}$, $V_{\text{SB}} = 0$) in the IHP SG13 130 nm CMOS technology, using the foundry's PSP models. The output characteristics show the triode and saturation regions including channel-length modulation, while the logarithmic transfer characteristics clearly show the exponential subthreshold region and the drain-current dependence on V_{DS} caused by DIBL.

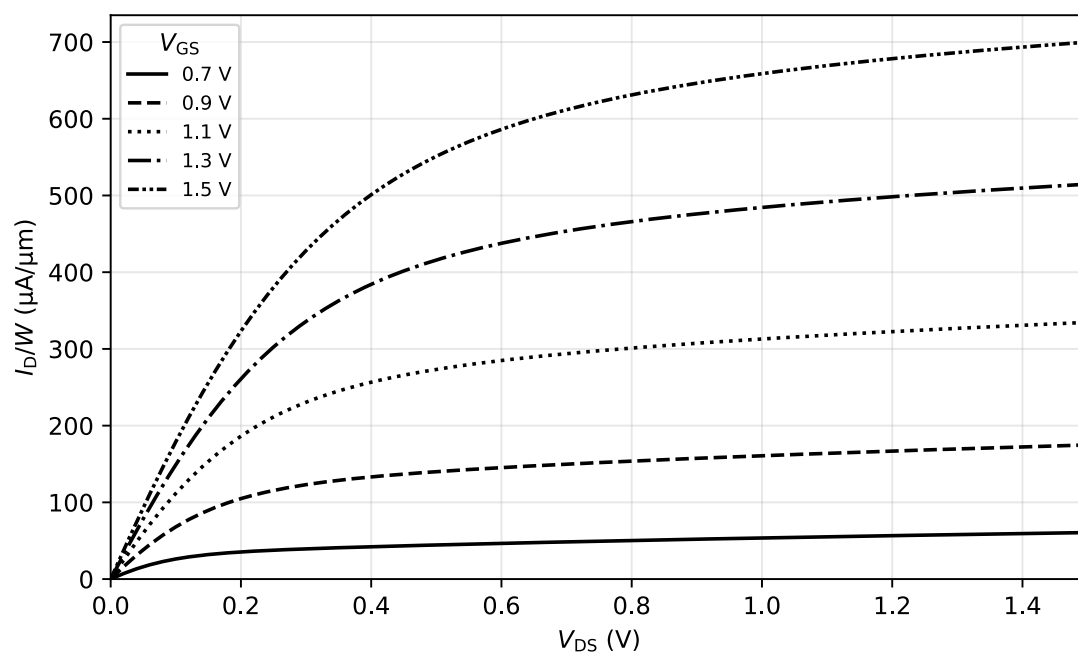


Figure 19: Simulated output characteristic of an NMOS in IHP SG13 ($W = 1.3 \mu\text{m}$, $L = 0.13 \mu\text{m}$, $V_{\text{SB}} = 0$) for different gate-source voltages.

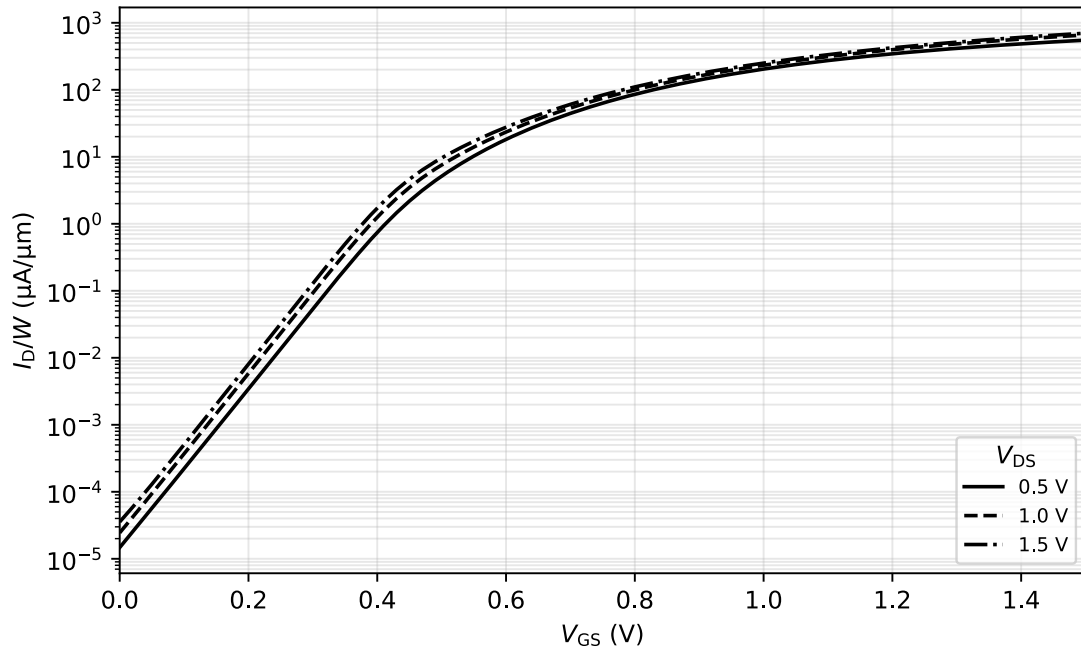


Figure 20: Simulated transfer characteristic of an NMOS in IHP SG13 ($W = 1.3 \mu\text{m}$, $L = 0.13 \mu\text{m}$, $V_{\text{SB}} = 0$) on a logarithmic current axis. The exponential subthreshold region is clearly visible, as is the drain-voltage dependence caused by DIBL.

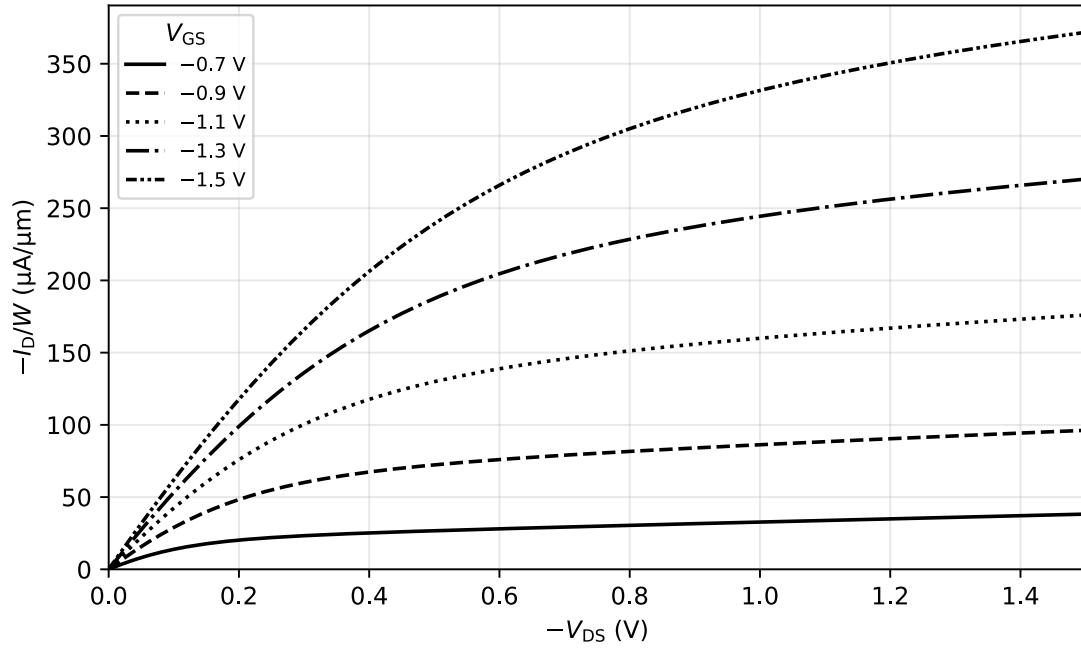


Figure 21: Simulated output characteristic of a PMOS in IHP SG13 ($W = 1.3 \mu\text{m}$, $L = 0.13 \mu\text{m}$, $V_{\text{SB}} = 0$) for different gate-source voltages. Note the lower current level compared to the NMOS due to the lower hole mobility.

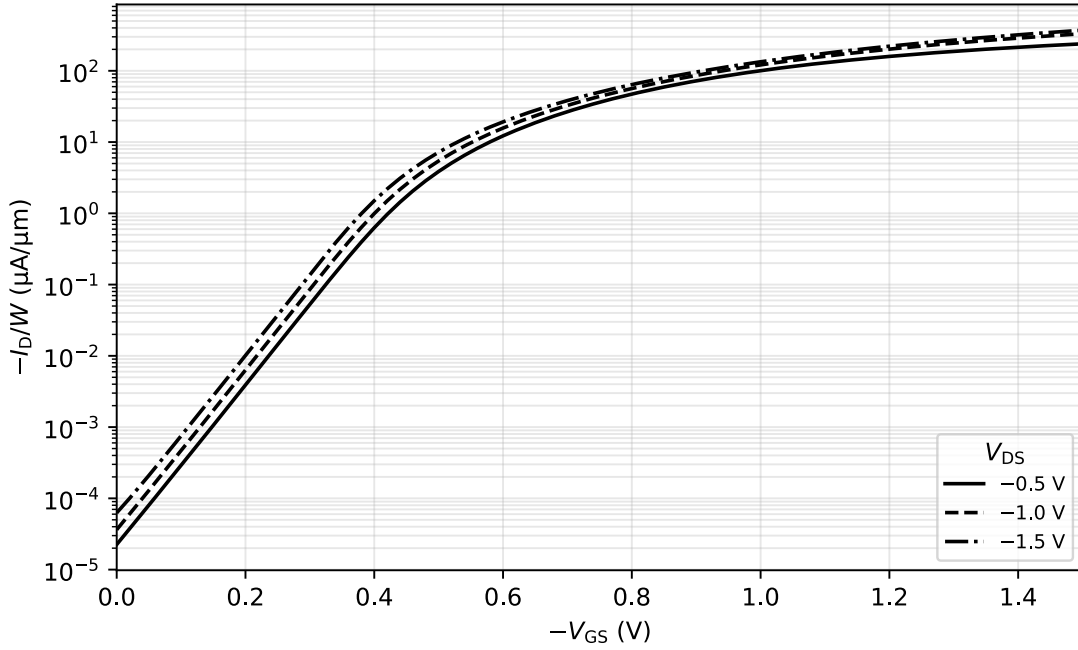


Figure 22: Simulated transfer characteristic of a PMOS in IHP SG13 ($W = 1.3 \mu\text{m}$, $L = 0.13 \mu\text{m}$, $V_{\text{SB}} = 0$) on a logarithmic current axis.

2.9 MOSFET Sizing

2.9.1 The g_m/I_D Method

When designing a circuit with MOSFETs, how should one select the parameters like I_D , V_{GS} , W , and L ? For digital designs, this is relatively straightforward: L is (usually) set to the minimum length, W is sized according to the driving requirements of the logic circuit, V_{GS} swings between V_{SS} and V_{DD} , and no static drain current flows. For analog, mixed-signal, and RF designs, the sizing question is more involved.

In the Analog Circuit Design course, the g_m/I_D method is used to find the proper sizing of a MOSFET depending on the circuit requirements. This method is the “go-to” method, as it uses numerical data found by evaluating the PDK models in a circuit simulator, and it describes the MOSFET correctly in all operating regions.

2.9.2 The Inversion-Coefficient Method

Despite the practical advantages of the g_m/I_D method, alternative approaches exist, and they can be helpful by providing relatively simple and unified analytical equations which are valid in all MOSFET operating regions. One such method is based on the **inversion coefficient**, which has been popularized as a design methodology by W. Sansen [10], [11]. Let us define a specific current

$$I_S = \mu C'_{\text{ox}} \frac{W}{L} \cdot 2nV_T^2. \quad (40)$$

Then, the inversion coefficient IC is defined as

$$\text{IC} = \frac{I_D}{I_S} = \left[\ln \left(1 + e^{\frac{V_{\text{GS}} - V_{\text{th}}}{2nV_T}} \right) \right]^2 \quad (41)$$

The beauty of this approach is that these equations work all the way from weak inversion (subthreshold operation, $IC < 0.1$) through moderate inversion ($IC \approx 1$) to strong inversion ($IC > 10$). Based on the selected IC , the important device properties can be calculated. The drain current is

$$I_D = I_S \cdot IC \quad (42)$$

the transconductance is given by

$$g_m = \frac{I_D}{nV_T} \frac{1 - e^{-\sqrt{IC}}}{\sqrt{IC}} \quad (43)$$

and the speed of the MOSFET is

$$f_T = \frac{1}{2\pi} \frac{3}{2} \frac{\mu}{L^2} V_T (\sqrt{1 + 4IC} - 1) \quad (44)$$

Let us check the asymptotic behavior of these equations against the earlier results (see Table 4).

Table 4: Asymptotic behavior of the inversion-coefficient equations in weak and strong inversion

Equa- tion	$V_{GS} \ll V_{th}$ (weak inversion)	$V_{GS} \gg V_{th}$ (strong inversion)
Equa- tion 41	$IC = e^{\frac{V_{GS}-V_{th}}{nV_T}}$	$IC = \left(\frac{V_{GS}-V_{th}}{2nV_T}\right)^2$
Equa- tion 42	$I_D = \mu C'_{ox} \frac{W}{L} 2nV_T^2 e^{\frac{V_{GS}-V_{th}}{nV_T}}$	$I_D = \frac{1}{2n} \mu C'_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$
Equa- tion 43	$g_m = \frac{I_D}{nV_T}$	$g_m = \frac{2I_D}{V_{GS}-V_{th}}$
Equa- tion 44	$f_T = \frac{1}{2\pi} \frac{3}{2} \frac{\mu}{L^2} 2V_T e^{\frac{V_{GS}-V_{th}}{nV_T}}$	$f_T = \frac{1}{2\pi} \frac{3}{2n} \frac{\mu}{L^2} (V_{GS} - V_{th})$

The asymptotes match the subthreshold model (Equation 24) and the corrected square-law model (Equation 28), respectively.

Plotting Equation 43 and Equation 44 versus IC , as done in Figure 23, reveals the classical trade-off between g_m -efficiency (g_m/I_D) and speed (f_T). Note that velocity saturation is not represented in this simple model, so f_T does not saturate at high IC in the plot.

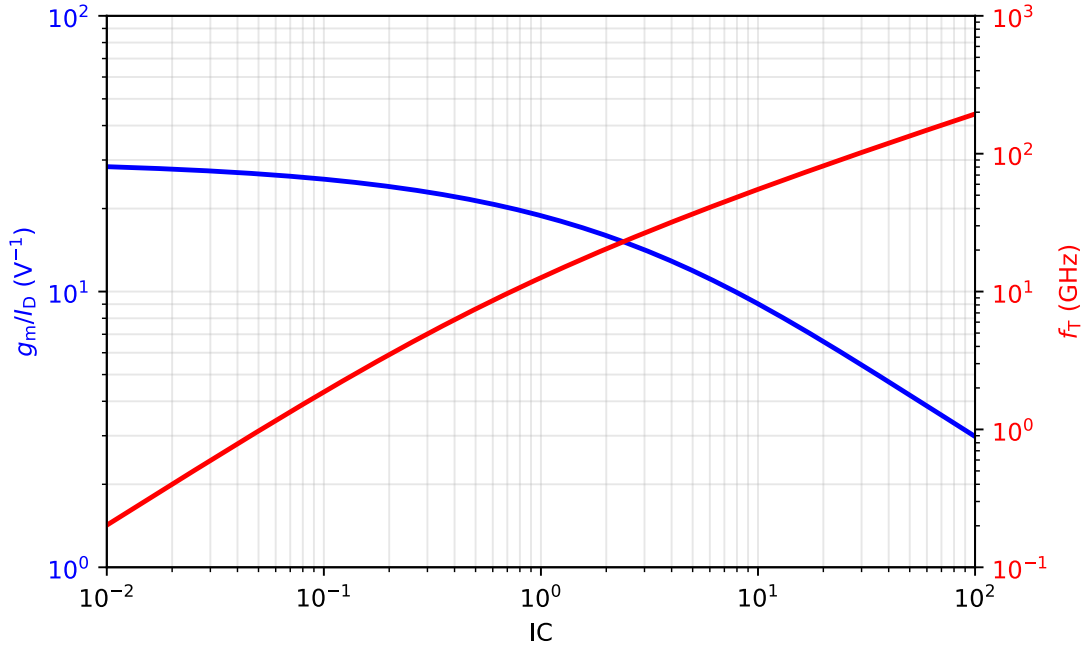


Figure 23: Transconductance efficiency g_m/I_D (blue, left axis) and transit frequency f_T (red, right axis) versus the inversion coefficient IC, calculated with the analytical model ($n = 1.3$, $L = 0.13 \mu\text{m}$, $\mu = 280 \text{ cm}^2/(\text{V s})$). Weak inversion offers the best efficiency, strong inversion the highest speed.

Figure 24 shows the same trade-off extracted from simulation data of an NMOS ($W/L = 1.3 \mu\text{m}/0.13 \mu\text{m}$) in IHP SG13. The simulation matches the analytical model well for g_m/I_D ; the simulated f_T comes out higher than the simple model predicts, since the model pessimistically assumes the full $\frac{2}{3}C'_{\text{ox}}WL$ gate capacitance at all bias points, while at high IC velocity saturation causes the simulated f_T to flatten.

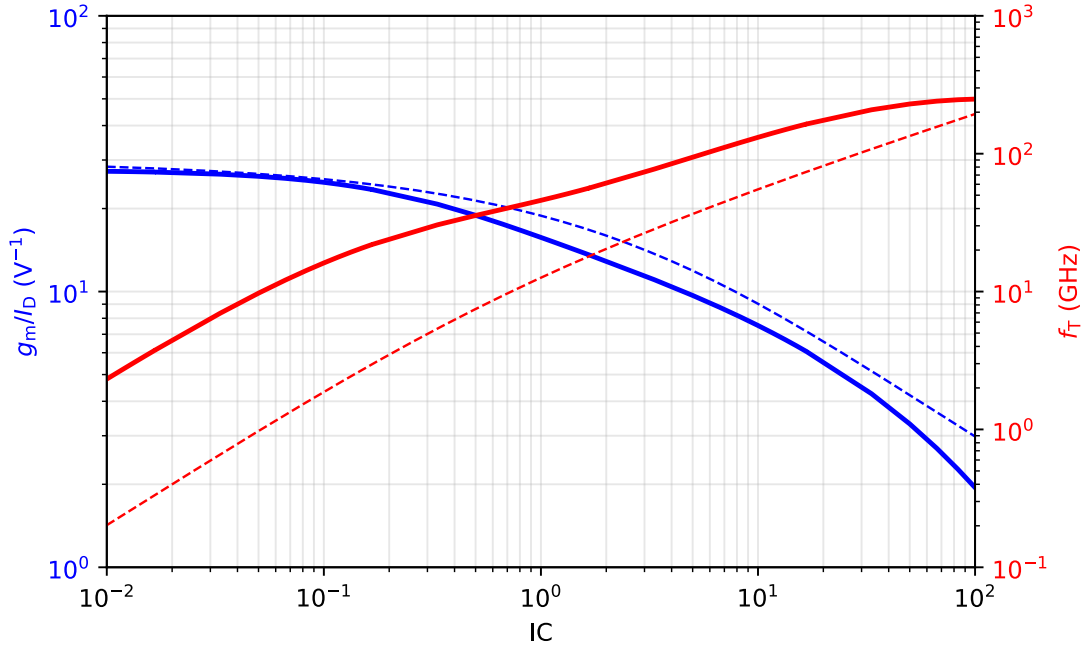


Figure 24: Transconductance efficiency g_m/I_D (blue, left axis) and transit frequency $f_T = g_m/(2\pi C_{gg})$ (red, right axis) versus inversion coefficient, extracted from a simulation of an NMOS ($W/L = 1.3 \mu\text{m}/0.13 \mu\text{m}$) in IHP SG13, using a specific current of $I_S = 6 \mu\text{A}$ extracted from the simulation data. The thin dashed lines show the analytical model (Equation 43 and Equation 44) with $n = 1.3$, $L = 0.13 \mu\text{m}$, and $\mu = 280 \text{ cm}^2/(\text{V s})$.

2.9.3 Sizing Procedure

The design procedure using the inversion coefficient is as follows:

1. Depending on the trade-off between low power (high g_m/I_D) and high speed (high f_T), select the IC (often, $IC \approx 1$ is a reasonable compromise).
2. Choose either the bias current or the required g_m , and calculate the other one using Equation 43.
3. Using Equation 42, calculate I_S from the known I_D .
4. Depending on the requirements for transistor speed (minimum L), matching (large $W \cdot L$), or small g_{ds} (large L), choose the appropriate L ; use Equation 40 to calculate the corresponding W .
5. Finally, estimate f_T using Equation 44. From f_T , the required bandwidth of the circuit can be estimated, and the sizing can be iterated if necessary.

Now the most important transistor parameters (W , L , I_D , g_m , V_{GS} , and f_T) are selected and known.

2.10 Summary

The MOSFET is a very versatile device, which can, depending on the bias conditions, work as a:

- **Switch** (for small V_{DS} and large V_{GS}).
- **Voltage-controlled resistor** (since $g_{ds} = f(V_{GS})$ in the triode region).
- **Voltage-controlled current source** (since $I_D = f(V_{GS})$ in the saturation region).

- **Diode** (since the MOSFET is off for $V_{GS} < V_{th}$ and on otherwise).
- **Variable capacitor** (using the gate oxide or drain/source junction capacitance).

This versatility is nicely illustrated by the collection of two-transistor circuits in [12]; the paper, plus posters showing all fifty circuits, is available at the Fifty Nifty Variations of Two-Transistor Circuits homepage.

When making the MOSFET physically small by scaling $L \ll 1 \mu\text{m}$ to improve power, performance, and area, various imperfections kick in (like DIBL), causing increased leakage currents and degraded output conductance—so alternative device constructions like SOI, FinFET, and nanosheet MOSFETs have been introduced.

2.11 Appendix: Silicon Material Properties

Table 5 collects the physical properties of silicon used throughout this text.

Table 5: Physical material properties of silicon

Physical parameter of Si	Symbol	Typical value
Lattice constant		0.357 nm
Density		$2.33 \times 10^3 \text{ kg/m}^3$
Relative permittivity of Si	$\epsilon_{r,\text{si}}$	11.9
Relative permittivity of SiO_2	$\epsilon_{r,\text{ox}}$	3.9
Band gap at 300 K	E_{bg}	1.12 eV
Intrinsic carrier concentration at 300 K	$n_i = p_i$	$1.0 \times 10^{10} \text{ cm}^{-3}$
Linear coefficient of thermal expansion	CTE	$2.6 \times 10^{-6} \text{ K}^{-1}$
Melting point		1415 °C
Mobility of electrons at 300 K	μ_n	$1500 \text{ cm}^2/(\text{V s})$
Mobility of holes at 300 K	μ_p	$450 \text{ cm}^2/(\text{V s})$
Thermal conductivity		$1.5 \text{ W}/(\text{cm K})$
Breakdown field	E_{crit}	$3 \times 10^5 \text{ V/cm}$

3 IC Technology

This chapter walks through the manufacturing technology of integrated circuits: from the raw silicon wafer through the individual processing steps (doping, photolithography, etching, thin-film deposition, and interconnect formation) to a complete CMOS process flow. The treatment largely follows [4].

3.1 The Silicon Wafer

The starting material for integrated-circuit production is, quite literally, sand: quartz sand (SiO_2) is reduced with coal in an electric arc furnace at about 2000 °C to roughly 98 % purity, following the reaction $\text{SiO}_2 + 2 \text{C} \rightarrow \text{Si} + 2 \text{CO}$. This “metallurgical-grade” silicon is then chemically purified via trichlorosilane ($\text{Si} + 3 \text{HCl} \rightleftharpoons \text{SiHCl}_3 + \text{H}_2$), which can be distilled and decomposed back into ultra-pure silicon.

From the purified silicon, a large single crystal—the **ingot**—is grown by the **Czochralski method**, illustrated in Figure 25: a seed crystal is dipped into molten silicon and slowly pulled upwards while rotating, so that the melt crystallizes at the interface, continuing the crystal structure of the seed. Further purification can be achieved by zone refining, where a small molten zone is moved along the ingot, dragging impurities along; the final purity reaches 99.999999 %, which is equivalent to one impurity atom per billion silicon atoms.

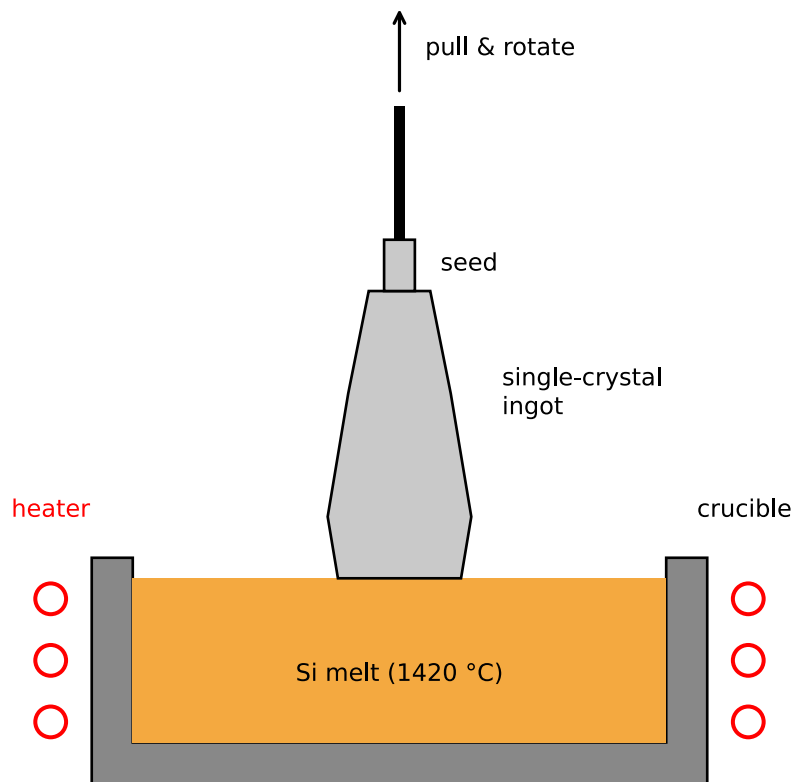


Figure 25: Czochralski crystal growth: a rotating seed crystal is slowly pulled out of the silicon melt, growing a large single-crystal ingot.

The ingot is then cut into **wafers** using a saw, to a thickness of 0.3 to 0.8 mm, and the wafers are polished. The standard wafer diameters today are 150 mm (6"), 200 mm (8"), and 300 mm (12"). The resistivity of the wafer material, set by doping, is 0.1 to 20 Ω cm, with up to 1000 Ω cm for specialty processes. Wafers can be *n*- or *p*-type, and often epitaxial ("epi") wafers are used.

3.1.1 Doping

The silicon atoms are spaced roughly 0.24 nm apart in the crystal lattice, resulting in about 5×10^{22} atoms/cm³. The intrinsic carrier concentration is $n_i = p_i \approx 1.0 \times 10^{10}$ cm⁻³ at 300 K, with an exponential temperature dependence (older textbooks often quote 1.5×10^{10} cm⁻³; the accepted measured value is 9.65×10^9 cm⁻³ [13]). By introducing doping materials in small quantities, the carrier concentration can be modulated strongly; see Table 6 (N_d and N_a are given in cm⁻³).

Table 6: Overview of n -type and p -type doping of silicon

n -type (donor materials)	p -type (acceptor materials)
Effect: Introduces additional free electrons	Introduces a lack of electrons ("holes")
Elements: Phosphorus (P), Arsenic (As)	Boron (B), Gallium (Ga), Indium (In)
Carrier concentrations: $n_0 \approx N_d, p_0 \approx n_i^2/N_d$	$p_0 \approx N_a, n_0 \approx n_i^2/N_a$

The resulting resistivity of doped silicon is shown in Figure 26 as a function of the doping concentration.

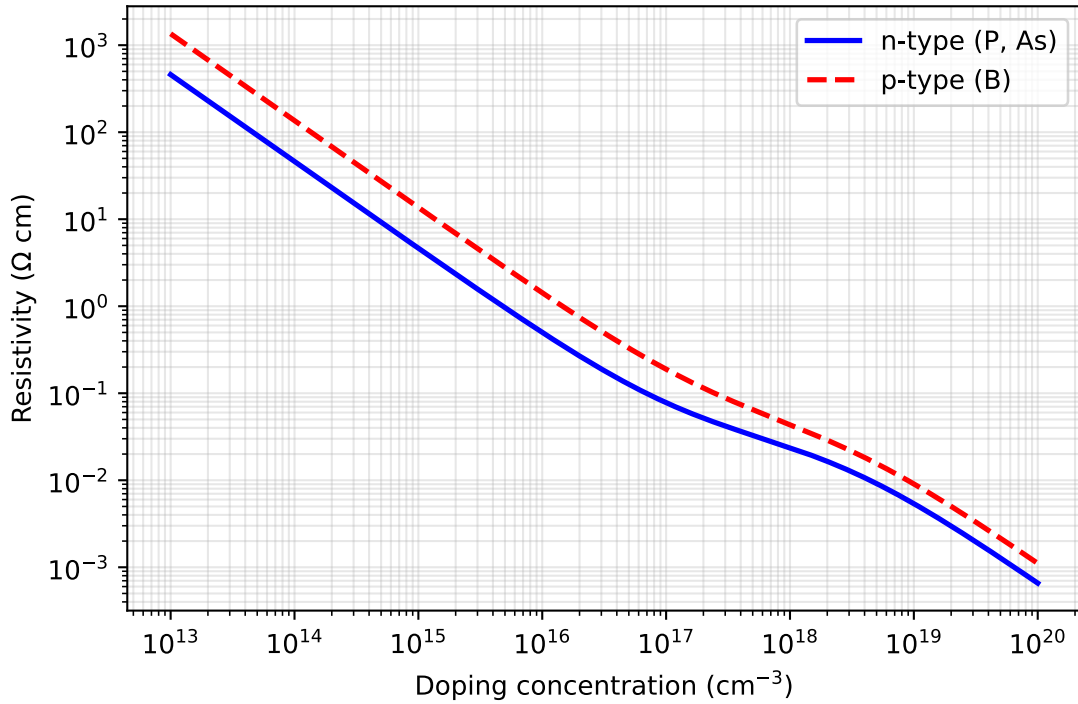


Figure 26: Resistivity of silicon at 300 K as a function of the doping concentration, calculated as $\rho = 1/(q\mu N)$ with doping-dependent carrier mobilities (Arora mobility fits). p -type silicon has a higher resistivity for the same doping level because of the lower hole mobility.

i Note 3: Example: Wafer Doping

Let us assume a p -type wafer with $50\ \Omega\text{ cm}$ resistivity (as used in IHP SG13 CMOS). Using Figure 26, we find $N_a = p_0 \approx 2.7 \times 10^{14}\text{ cm}^{-3}$, which results in $n_0 \approx n_i^2/N_a = 3.7 \times 10^5\text{ cm}^{-3}$. This doping concentration equates to about one acceptor atom per 180 million silicon atoms!

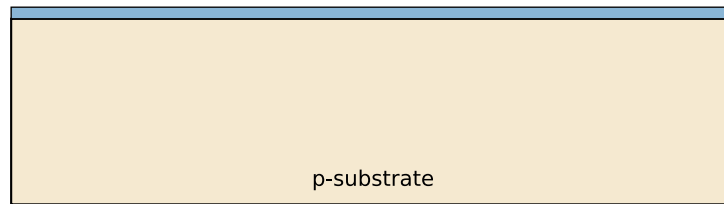
3.2 Selective Doping

To introduce dopants into the silicon wafer selectively, a masking process is used:

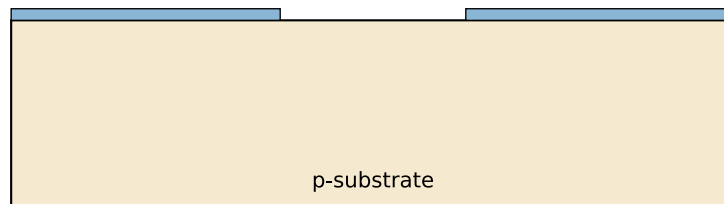
- The picture from a photomask is optically transferred into a photoresist on the wafer, using a process called **photolithography**.
- This photoresist is developed and used to selectively etch a masking layer, like SiO_2 (a “hard mask”).
- After this step, the silicon surface is opened at specific locations and covered otherwise, allowing the introduction of dopants into defined areas of the wafer.

The complete sequence—hard-mask formation, lithography, etching, dopant introduction, and annealing (which removes defects from the crystal structure while the dopants diffuse into the silicon)—is shown in Figure 27.

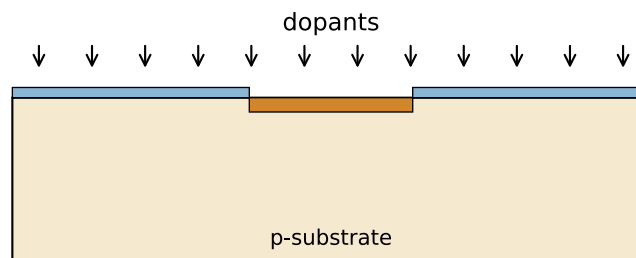
(a) grow/deposit hard mask



(b) open hard mask by lithography and etching



(c) introduce dopants



(d) anneal (dopants diffuse into the Si)

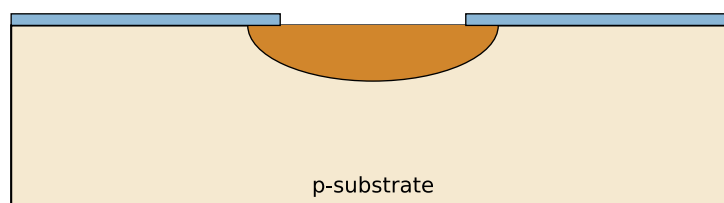


Figure 27: Selective doping using a hard mask: (a) the wafer is covered by a hard mask (e.g., SiO_2), (b) the hard mask is opened by photolithography and etching, (c) dopants are introduced through the opening (shown here by implantation), and (d) after annealing, the dopants have diffused into the silicon and the crystal damage is repaired.

3.2.1 Oxidation of Silicon

Oxidation of silicon is done at elevated temperatures in a furnace, using either “dry” ($\text{Si} + \text{O}_2 \rightarrow \text{SiO}_2$) or “wet” ($\text{Si} + 2 \text{H}_2\text{O} \rightarrow \text{SiO}_2 + 2 \text{H}_2$) oxidation. The oxide growth follows the **Deal-Grove model** [14]; the resulting growth curves are shown in Figure 28. Wet oxidation

is much faster, while dry oxidation yields denser, higher-quality oxides (used, e.g., for gate oxides).

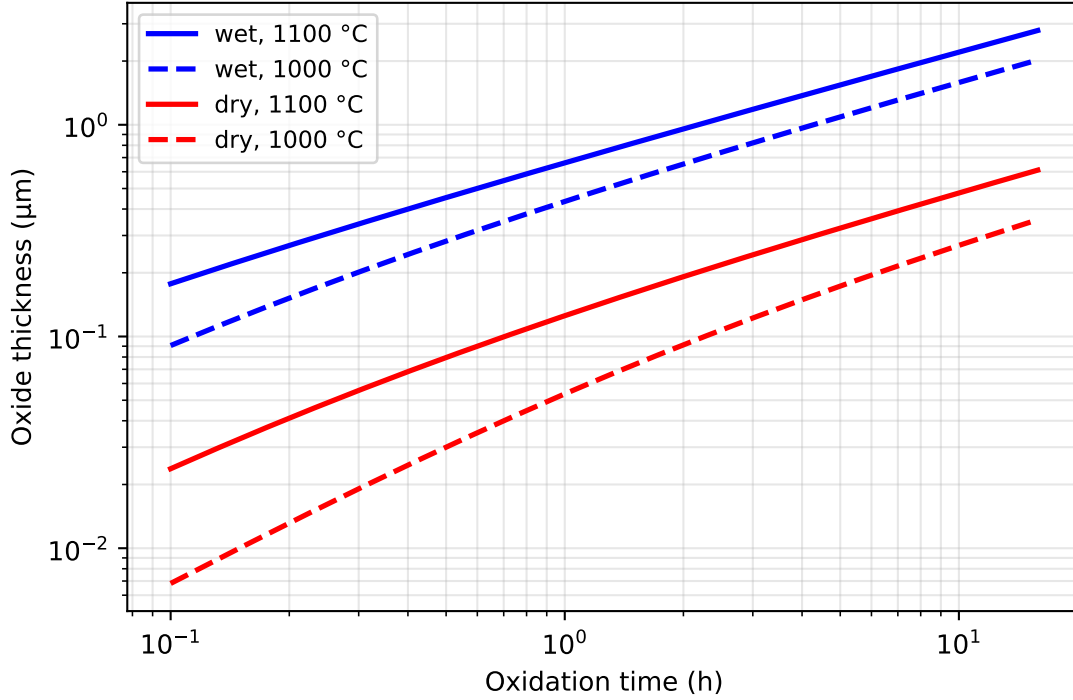


Figure 28: Thermal oxide growth on silicon calculated with the Deal-Grove model, $x_{\text{ox}}^2 + Ax_{\text{ox}} = Bt$, for wet and dry oxidation at different temperatures. Wet oxidation is much faster and used for thick isolation oxides, dry oxidation for thin, high-quality gate oxides.

The SiO_2 is used for:

- The gate dielectric of the MOSFET.
- Hard masks for implantation or doping.
- Device isolation and isolation of wiring.

The growth of SiO_2 consumes silicon; the silicon thickness is reduced by 44 % of the grown oxide thickness. Alternatively, chemical vapour deposition (CVD) can be used for depositing SiO_2 without consuming silicon. Local oxidation of silicon (LOCOS) can be used to laterally isolate active regions of an integrated circuit, but in modern CMOS technologies shallow trench isolation (STI) is used instead, for higher transistor density.

3.3 Photolithography

In photolithography, the wafer is first oxidized to create a hard-mask layer and then coated with photoresist. The photomask is imaged onto the photoresist with a projection lens system, the resist is developed and baked, and the hard mask is etched through the resist openings. There are two types of resist: with a **positive** resist, the exposed areas dissolve in the developer, transferring the mask image directly; with a **negative** resist, the exposed areas remain, creating an inverse image. After etching, the remaining photoresist is removed. The process is illustrated in Figure 29.

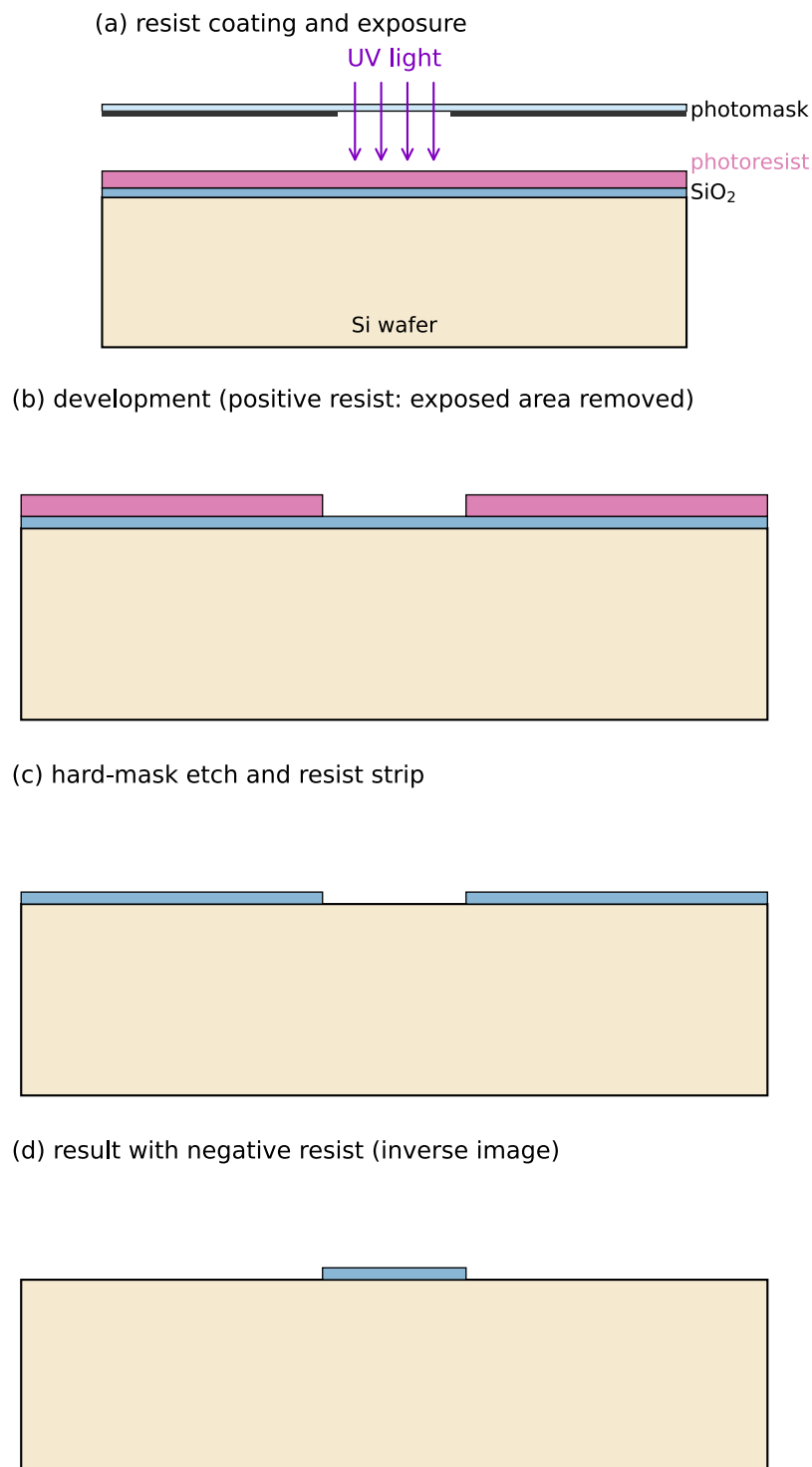


Figure 29: Patterning by photolithography: (a) the oxidized wafer is coated with photoresist and exposed through the photomask, (b) with a positive resist, the exposed resist dissolves during development, (c) the hard mask is etched through the resist opening and the resist is removed, and (d) a negative resist creates the inverse image.

3.3.1 Photomasks

Instead of a photomask which exposes the whole wafer at once, a **reticle** is mostly used, which is smaller than the wafer, with a field size of ca. 26 mm × 33 mm (limiting the maximum die size to 858 mm²). A **stepper** exposes the complete wafer field by field with the reticle. The photomask/reticle is usually made of a glass carrier with a structured chromium layer.

The reticle must be protected from contamination, so a thin transparent membrane (called a **pellicle**) keeps particles out of the focal plane, away from the mask structures. The reticle can be the same size (1×) or larger (2× or 4×) than the structures on the wafer, and the photomask can touch the wafer (“contact printing”) or be kept at a distance (“proximity printing”)—modern lithography uses projection printing through a reduction lens.

The complete exposure system is shown in Figure 30. The light passes from the source through the **illuminator**, which shapes the angular distribution of the illumination (e.g. into a dipole or quasar pattern), onto the reticle; the projection optics then image a demagnified copy of the reticle field into the resist, and the scanner steps this field across the wafer, one exposure at a time.

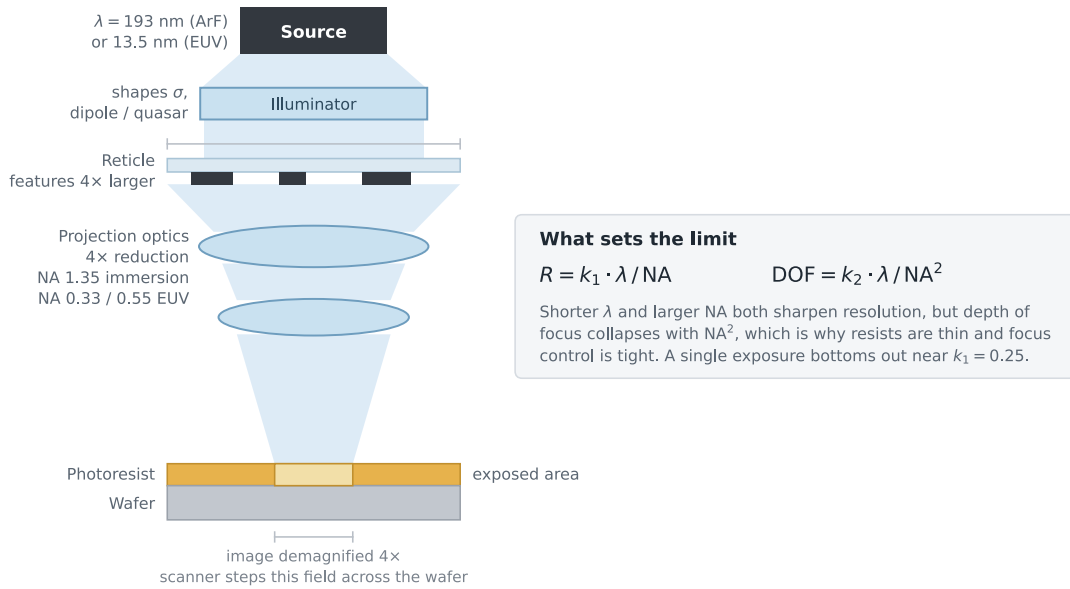


Figure 30: The optical column of a projection lithography scanner: light from the source passes through the illuminator, which shapes its angular distribution, onto the reticle, and the projection optics image a demagnified copy of the reticle field into the photoresist. The scanner exposes the wafer field by field. Resolution and depth of focus both follow from the wavelength and the numerical aperture of the optics.

3.3.2 Resolution Limit

As in any other optical system, the resolution limit of photolithography is given by the Rayleigh criterion:

$$R = k_1 \frac{\lambda}{n \sin \Theta} = k_1 \frac{\lambda}{NA} \quad (45)$$

Here, λ is the wavelength of the light source, and k_1 is a factor that captures various resolution-enhancement measures, like phase-shift masks or off-axis illumination ($0.25 \leq k_1 \leq 1$). NA is the numerical aperture of the lens system (in air, $\text{NA} \leq 0.93$). The resulting half-pitch R is the smallest pattern that can be printed on a wafer: $R = 100$ nm means that lines with a width of 100 nm and a spacing of 100 nm can be printed.

Resolution is not free. The **depth of focus**, i.e. the range over which the projected image stays sharp, shrinks with the *square* of the numerical aperture:

$$\text{DOF} = k_2 \frac{n\lambda}{\text{NA}^2} \quad (46)$$

Here, k_2 is the depth-of-focus counterpart of k_1 : it captures how much image degradation the process still tolerates, set by resist contrast and thickness, feature type and illumination scheme ($0.5 \leq k_2 \leq 1$). The two factors pull against each other—the same off-axis illumination that buys a small k_1 also shrinks k_2 , so pushing resolution costs focus budget twice. Note that the refractive index n does not cancel here: raising n recovers depth of focus at a given NA.

This is why resists are kept thin and why the wafer surface must be held in focus to within a few tens of nanometers while the field is scanned.

3.3.3 Light Sources

In the 1980s, mercury lamps with $\lambda = 436$ nm (g-line) or $\lambda = 365$ nm (i-line) were used. Starting with 0.25 μm CMOS (until 180 nm), excimer lasers with $\lambda = 248$ nm were introduced, called deep-ultraviolet (DUV) light. Since 130 nm CMOS, ArF lasers with $\lambda = 193$ nm are used, which can be combined with multi-patterning to break the half-pitch limit of $R \approx 40$ nm. Finally, extreme-ultraviolet (EUV) lithography was introduced at the 7 nm CMOS node, offering a huge reduction in wavelength to $\lambda = 13.5$ nm—accompanied by an extreme cost impact due to the need for all-reflective optics.

Note that there is only one company in the world that currently manufactures EUV lithography machines: ASML from the Netherlands. This gives ASML a unique position in the semiconductor equipment market, as EUV is critical for advanced nodes. Zeiss from Germany produces the high-precision optics required for these machines. Further, there is only one company in the world that makes the multi-beam mask writers needed for creating the EUV masks: IMS Nanofabrication from Austria.

3.3.4 Resolution Enhancements

Various methods are used to reduce the half-pitch in Equation 45 by compensating distortion due to diffraction and interference, modifying either the amplitude or the phase of the light:

- **Optical proximity correction (OPC)** enhances the contrast of critical layout features like line ends and corners by pre-distorting the mask shapes.
- **Phase-shift masks (PSM)** vary the thickness of the mask, so that phase changes of the light can be used for contrast enhancement.
- **Immersion lithography** uses the refractive index of water ($n = 1.44$) between lens and wafer to achieve $\text{NA} > 1$; per Equation 46, the higher n also restores part of the depth of focus lost to the larger NA.

Ultimately, using these techniques, structures considerably smaller than the wavelength of the used light can be printed.

3.3.5 EUV Lithography

EUV light at 13.5 nm is created by firing laser pulses at molten tin droplets. There are no transmissive lens materials for EUV, so the optics are constructed from reflective mirrors with about 30 % loss per mirror. A first-generation EUV scanner is a 180-tonne machine using 0.5 MW of electrical power to create roughly 200 W of optical power at the wafer, processing about 200 wafers per hour. Comparing the achievable resolution:

- ArF immersion lithography: $\lambda = 193 \text{ nm}$, $\text{NA} = 1.35$, $k_1 = 0.27$ (with PSM) $\Rightarrow R \approx 40 \text{ nm}$.
- First-generation EUV (NXE): $\lambda = 13.5 \text{ nm}$, $\text{NA} = 0.33$, $k_1 = 0.32 \Rightarrow R \approx 13 \text{ nm}$.
- Current high-NA EUV (EXE): $\lambda = 13.5 \text{ nm}$, $\text{NA} = 0.55$, $k_1 = 0.32 \Rightarrow R \approx 8 \text{ nm}$.

3.4 Etching

Etching is used to transfer the image from the photoresist into the hard mask (or from one layer into another layer) by selectively removing material. Etching can be isotropic or anisotropic, characterized by the anisotropy A ; the selectivity S measures the ratio of the etch rate of the film to the etch rate of the mask. An optimal etch has $A = 1$ (only vertical etching, no lateral etch) and $S = \infty$ (the mask is not attacked). Two types of etches are used:

- **Wet etching**, using liquid chemicals (typically isotropic).
- **Dry etching**, using chemically active ionized gases in a plasma reactor (can be highly anisotropic).

3.5 Doping Techniques

Three methods of introducing dopants into the silicon can be used:

- **Diffusion**: dopants move (by diffusion) from the wafer surface into the wafer material; the source of the dopants is a gas (e.g., POCl_3 for phosphorus doping) or a solid film on the wafer surface.
- **Ion implantation**: dopants are ionized, accelerated, and shot into the wafer material.
- **Epitaxy**: additional silicon is grown on the wafer surface and doped in situ while growing the silicon layer.

3.5.1 Diffusion

Diffusion is the movement of impurity atoms from the surface into the bulk of the silicon, performed at high temperatures (900 to 1200 °C). There are two limiting cases:

- **Infinite source** of impurities at the surface: the resulting doping profile follows a complementary error function (erfc).
- **Finite source** of impurities at the surface: the resulting doping profile is a Gaussian, $N(x, t) = N_0 / \sqrt{\pi D t} \cdot \exp(-x^2 / (4 D t))$, with x the distance from the wafer surface, N_0 the initial dopant dose per area, D the diffusivity, and t the diffusion time.

The drawback of diffusion is that the doping concentration is always highest at the wafer surface and constrained to these profile shapes.

3.5.2 Ion Implantation

Ion implantation lodges impurity ions into the target material at high velocity; implantation can even be done through surface layers (e.g., through the gate oxide for threshold-voltage adjustment). The resulting doping depth profile is

$$N(x) = \frac{N_i}{\sqrt{2\pi} \Delta R} e^{-\frac{(x-R)^2}{2\Delta R^2}} \quad (47)$$

which peaks at the projected range R ; the straggle ΔR is tied to R , so the achievable profiles are also constrained (N_i is the implanted dose per area). A subsequent anneal (typically a rapid thermal anneal at 900 to 1100 °C) is required to activate the impurities and to repair the damage to the crystal lattice. Figure 31 compares typical diffusion and implantation profiles.

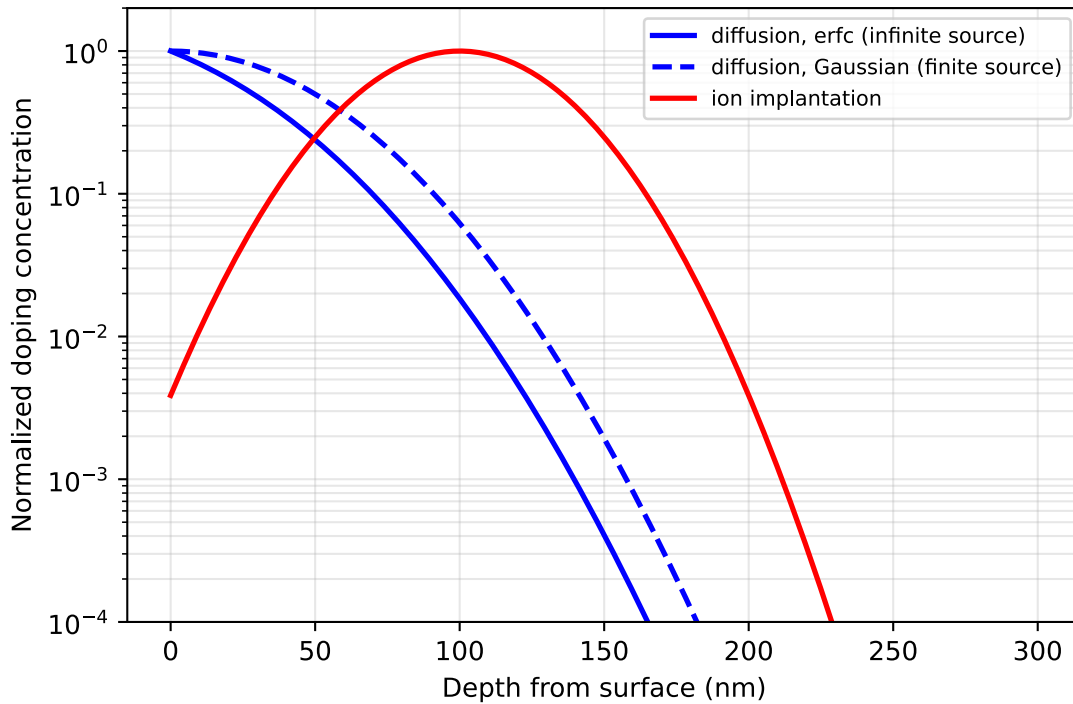


Figure 31: Doping depth profiles (normalized): diffusion always peaks at the wafer surface, following an erfc profile (infinite source) or a Gaussian profile (finite source), while ion implantation places the doping peak at the projected range R below the surface (shown for $R = 100$ nm, $\Delta R = 30$ nm).

3.5.3 Epitaxy

Epitaxial growth forms a layer of single-crystal silicon on the surface of the wafer, such that the crystal structure is continuous across the interface. The epi layer can be doped differently from the material where it is grown, with a typical thickness of 1 to 20 μm (patterned epitaxial growth using a mask is also possible).

3.6 Thin-Film Deposition

Various thin films are used in IC processing:

- The gate dielectric of the MOSFET, like SiO_2 (either thermally grown or deposited via CVD).
- The gate material, polysilicon, which is also used for local interconnects and resistors.

- Metal films for interconnects (metals like Al or Cu, and silicides like TiSi_2 , WSi_2 , TaSi_2 , NiSi_2).
- Contact and via materials, like W, Al, or Cu.
- Dielectrics between metal layers and spacers, like SiO_2 and Si_3N_4 .

These films are either **sputtered** or deposited from chemical gases in **chemical vapour deposition (CVD)**; both methods are sketched in Figure 32. Exemplary chemical reactions for CVD are:

- Depositing polysilicon: $\text{SiH}_4 \rightarrow \text{Si} + 2 \text{H}_2$
- Depositing SiO_2 (as an alternative to thermal oxidation): $\text{SiH}_4 + \text{O}_2 \rightarrow \text{SiO}_2 + 2 \text{H}_2$ or $\text{SiH}_2\text{Cl}_2 + 2 \text{N}_2\text{O} \rightarrow \text{SiO}_2 + 2 \text{HCl} + 2 \text{N}_2$
- Depositing Si_3N_4 : $3 \text{SiH}_2\text{Cl}_2 + 4 \text{NH}_3 \rightarrow \text{Si}_3\text{N}_4 + 6 \text{HCl} + 6 \text{H}_2$

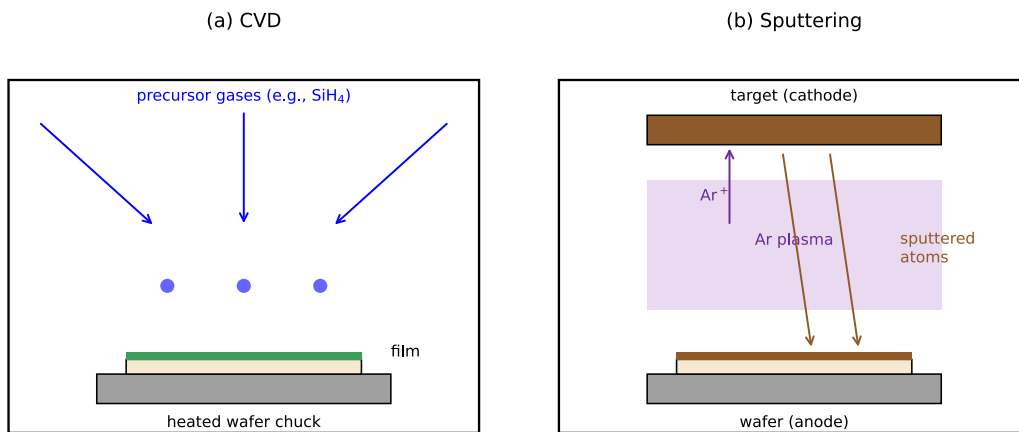


Figure 32: Thin-film deposition methods: (a) in chemical vapour deposition (CVD), precursor gases react at the heated wafer surface and deposit a thin film; (b) in sputtering, ions from a plasma knock atoms out of a target, which then condense on the wafer surface.

3.7 Interconnects

In the so-called **back-end** of the process, the devices formed in the **front-end** (like MOSFETs and resistors) are connected by a sandwich of metallic and dielectric thin films. The metal layers are connected through the dielectric layers by **vias** (between metals) and **contacts** (from the lowest metal layer to the front-end devices).

Historically, aluminum has been used as the interconnect metal, as it can be deposited by sputtering and then patterned by etching. However, Al is prone to void formation by electromigration. In newer technologies, copper is used: its resistance is 40 % lower, and it has excellent electromigration reliability. However, dry etching of Cu is difficult, so a **damascene** process is used, shown in Figure 33: the dielectric is structured, a liner and Cu are deposited into the trenches, and the excess metal is removed by **chemical-mechanical polishing (CMP)**. The liner prevents the Cu from diffusing into the dielectric.

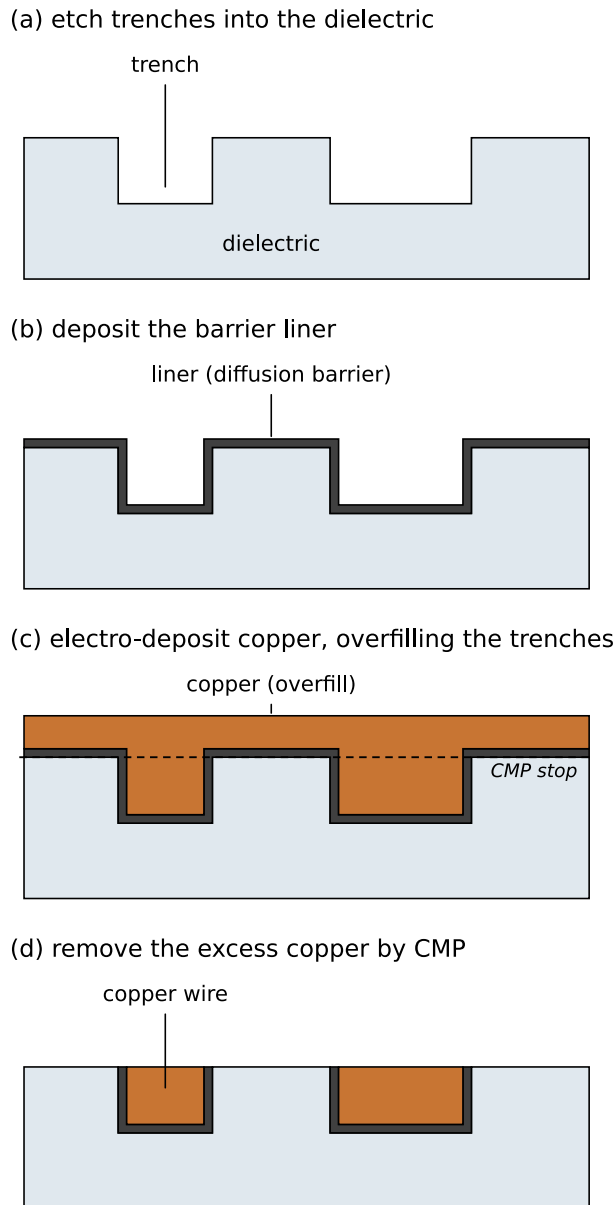


Figure 33: The damascene process for copper interconnects: (a) trenches are etched into the dielectric, (b) a conformal diffusion-barrier liner is deposited, (c) copper is (electro-)deposited, overfilling the trenches, and (d) the excess copper and the liner on the field are removed by chemical-mechanical polishing (CMP), leaving metal only in the trenches.

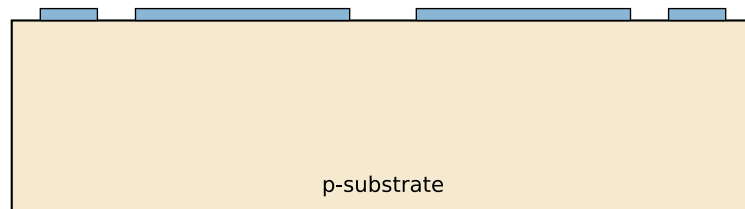
The classic dielectric between the metal layers is SiO_2 , but in order to reduce capacitive parasitics, low-k dielectrics (e.g., F- or C-doped SiO_2) are used in modern processes. Between processing steps, CMP is used to planarize the surfaces.

3.8 An Exemplary CMOS Process Flow

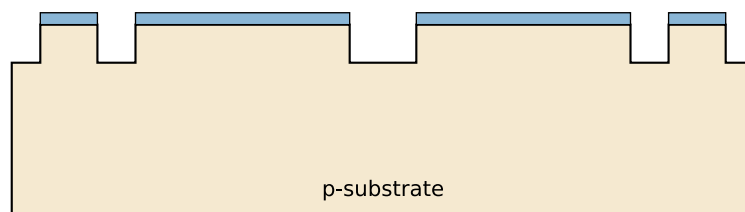
The following figures illustrate a simplified CMOS process flow, inspired by the openly documented LibreSilicon process; the cross sections show an NMOS (in the p-well) and a PMOS (in the n-well), each with its well tap on the outside, evolving through the process steps.

Step 1—Shallow trench isolation: After an initial cleaning, a hard mask (SiO_2 or Si_3N_4) is grown/deposited and patterned by photolithography. The isolation trenches are etched anisotropically and filled with oxide (Figure 34).

(a) hard mask grown and patterned



(b) anisotropic etch of the isolation trenches



(c) trenches filled with oxide, hard mask removed

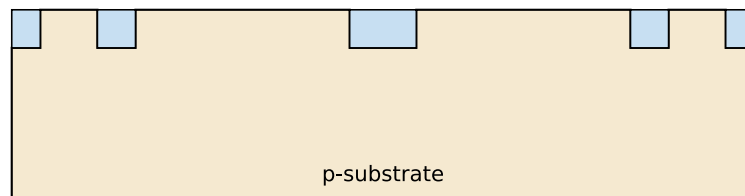


Figure 34: CMOS process, step 1—shallow trench isolation: (a) a hard mask is grown and patterned by photolithography, (b) the isolation trenches are etched anisotropically into the silicon, and (c) the trenches are filled with oxide and the surface is planarized.

Steps 2 and 3—Well formation: A hard mask is formed and opened over the p-well region, boron is implanted, and the wafer is annealed. The sequence is repeated with phosphorus for the n-well (Figure 35).

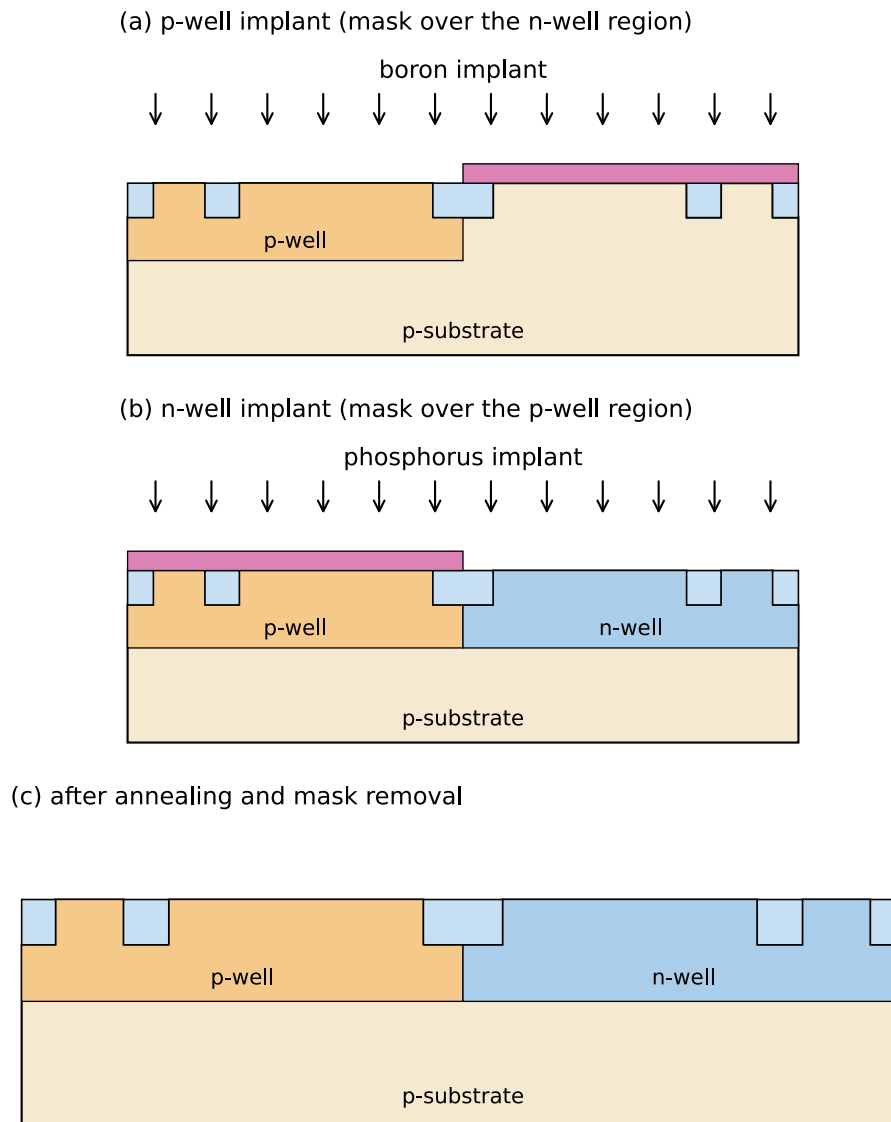
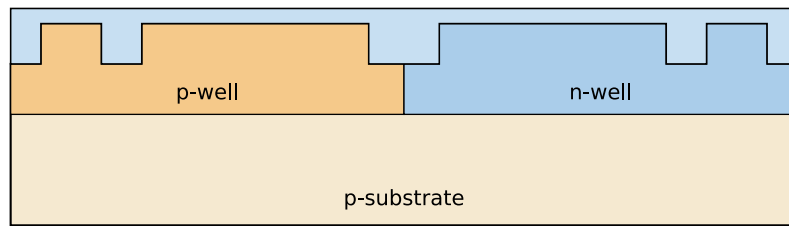


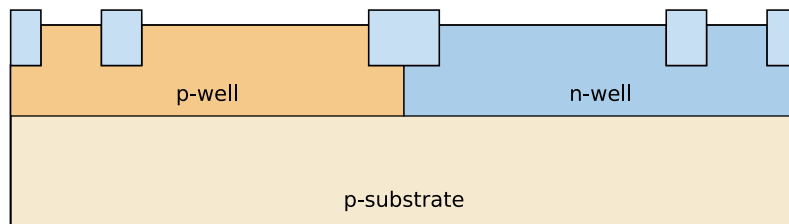
Figure 35: CMOS process, steps 2 and 3—well formation: (a) a mask covers the right half of the wafer while boron is implanted to form the p-well, (b) the complementary mask is used for the phosphorus implant of the n-well, and (c) after annealing, both wells are formed.

Steps 4 and 5—Gate formation: A field oxide is grown and removed over the active areas; then the thin gate oxide is grown by dry oxidation, polysilicon is deposited by CVD, and the gate stack is patterned (Figure 36).

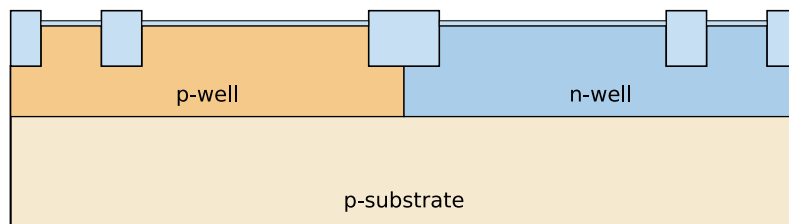
(a) field-oxide growth over the whole wafer



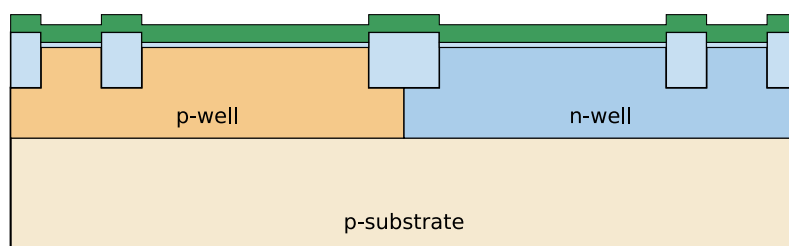
(b) field oxide removed over the active areas



(c) gate-oxide growth (dry oxidation)



(d) polysilicon deposition (CVD)



(e) gate stack patterned

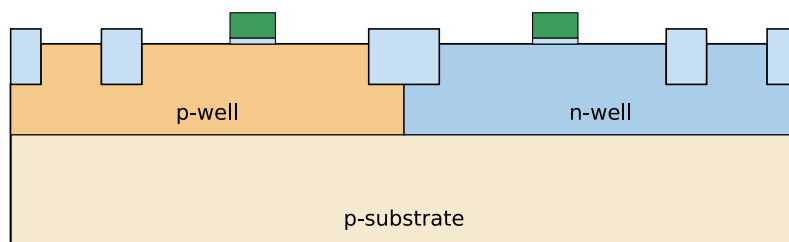


Figure 36: CMOS process, steps 4 and 5—gate formation: (a) the field oxide is grown over the whole wafer, merging with the oxide already filling the trenches, (b) it is removed again over the active areas, (c) the thin gate oxide is grown by dry oxidation on the exposed active areas, (d) polysilicon is deposited by CVD over the whole wafer, following the topography of the field oxide, and (e) the gate stack is patterned by photolithography and etching.

Steps 6 and 7—Source/drain implants: With photoresist masking the PMOS area, phosphorus is implanted to form the NMOS source/drain regions (and the n-well contacts); the implant is **self-aligned** to the gate, which acts as an implantation mask for the channel region. The complementary boron implant forms the PMOS source/drain regions (and the p-well contacts). Each implant is followed by an anneal (Figure 37).

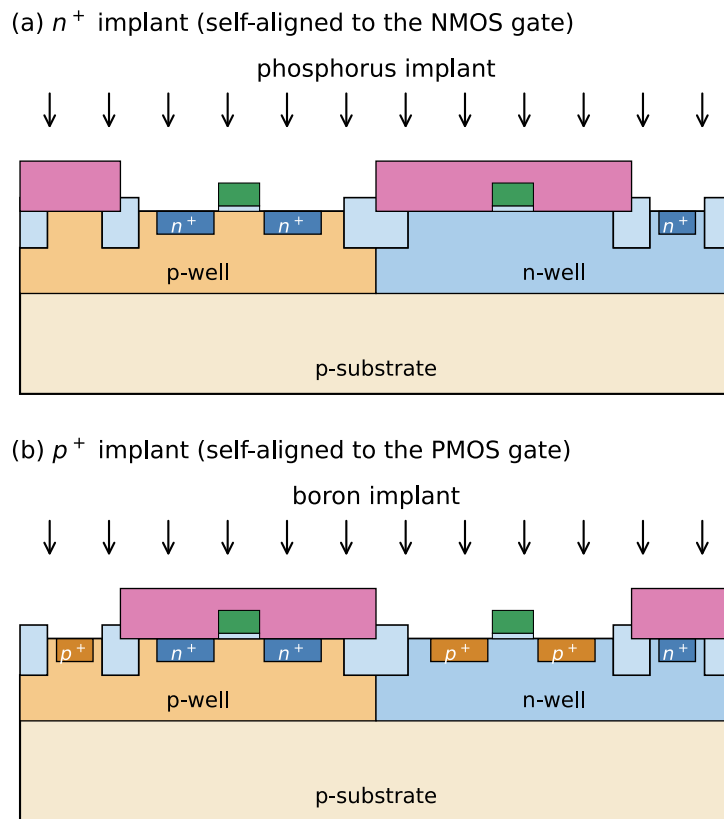
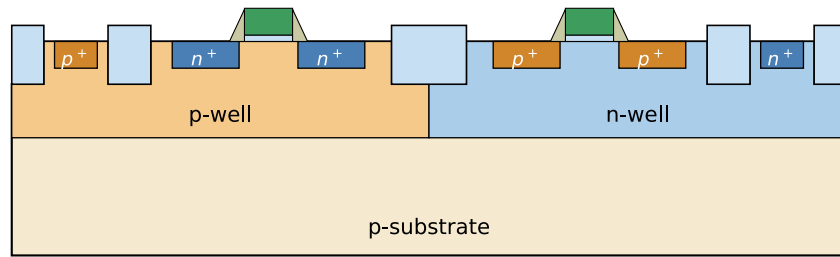


Figure 37: CMOS process, steps 6 and 7—self-aligned source/drain implants: (a) with the PMOS and the p-well tap covered by resist, phosphorus is implanted; the polysilicon gate itself masks the channel, so the n^+ regions are self-aligned to the gate, and the same implant forms the n^+ tap contacting the n-well; (b) the complementary boron implant forms the p^+ source/drain regions of the PMOS and the p^+ tap contacting the p-well.

Step 8—Silicide formation: An oxide/nitride layer is deposited and anisotropically etched, leaving **spacers** at the gate edges. Titanium is sputtered onto the wafer and reacts where it touches silicon, forming low-resistance TiSi_2 on the source/drain areas and the gate; the unreacted metal is removed (Figure 38).

(a) spacer formation at the gate edges



(b) silicide (TiSi_2) on source/drain and gate

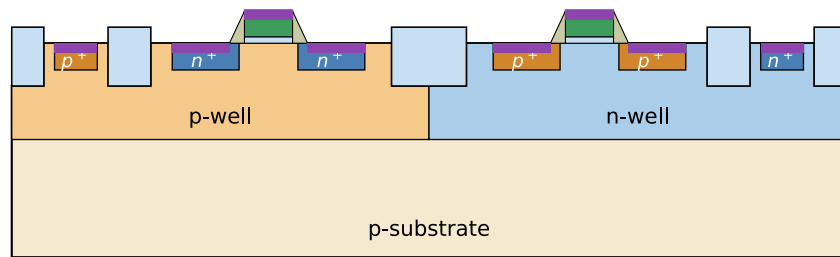
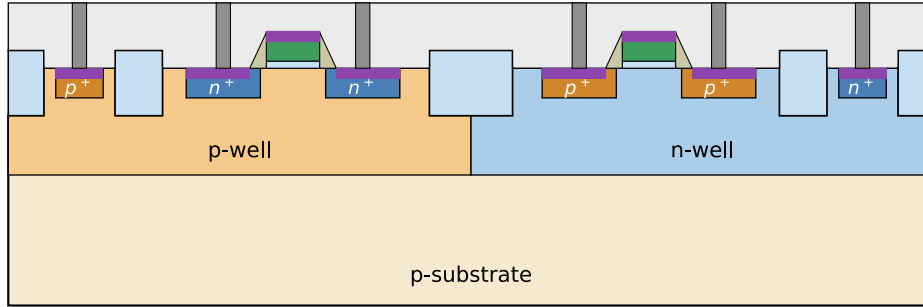


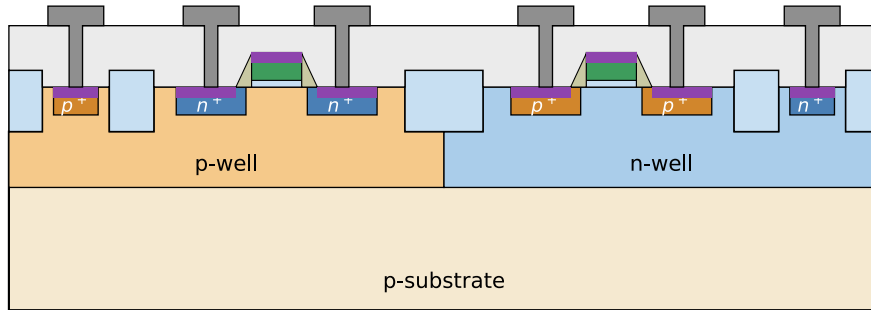
Figure 38: CMOS process, step 8—spacer and silicide formation: (a) an oxide layer is deposited and anisotropically etched, leaving spacers at the gate edges, and (b) sputtered titanium reacts where it touches silicon, forming a low-resistance silicide (dark) on source/drain and gate; the unreacted metal is then removed.

Steps 9 to 11—Contacts and metal stack: An inter-layer dielectric is deposited and planarized by CMP, contact holes are etched and filled, and the first metal layer is deposited and patterned. Further levels of dielectric, vias, and metal follow; finally, the passivation is deposited and opened at the pads (Figure 39).

(a) inter-layer dielectric, CMP, and contacts



(b) metal-1 sputtered (Al) and patterned



(c) inter-metal dielectric, via-12, and metal-2

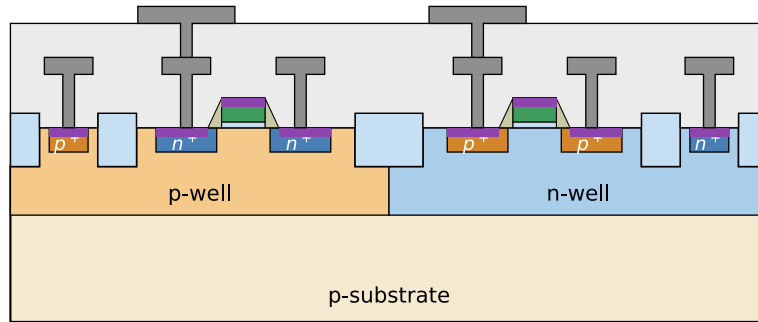


Figure 39: CMOS process, steps 9 to 11—back-end: (a) inter-layer dielectric deposition, CMP, and contact formation, (b) first metal layer sputtered and patterned, and (c) inter-metal dielectric, vias, and second metal layer; the finished wafer is then passivated and opened at the pads.

3.9 The Open-Source PDK IHP SG13

IHP has open-sourced the process design kit (PDK) for a 130 nm BiCMOS technology (the current generation is named SG13G2, referred to here simply as IHP SG13) under the Apache-2.0 license. The relevant files, including documentation, can be found at github.com/IHP-GmbH/IHP-Open-PDK. This technology has a lot of interesting features:

- Support for internal 1.2 V/1.5 V core MOSFETs and 3.3 V I/O MOSFETs

- An aluminum metal stack with 5 thin and 2 thick levels of metal (see Figure 40)
- Silicided, standard, and high-sheet-rho poly resistors
- A metal-insulator-metal (MIM) capacitor
- A Schottky diode
- Optional high-speed SiGe:C heterojunction bipolar transistors (HBTs)

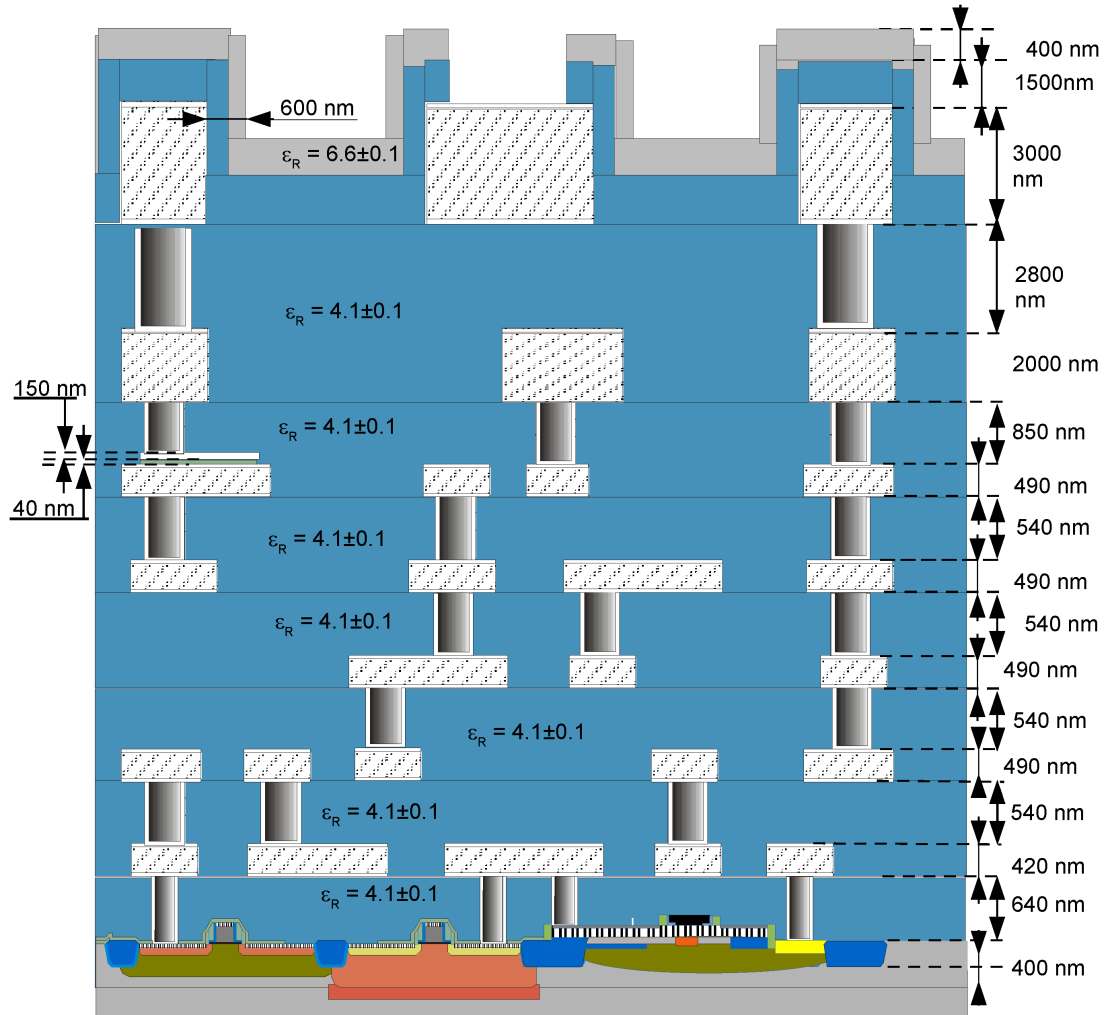


Figure 40: Cross-section of the IHP SG13 technology (not to scale): five thin metal layers (Metal1 to Metal5) for dense routing and two thick top metal layers (TopMetal1, TopMetal2) for power routing, inductors, and low-loss RF interconnects, with the MIM capacitor embedded between Metal5 and TopMetal1. The FEOL devices are shown at the bottom, and the numbers give the layer and dielectric thicknesses. Source: Fig. 1.1.1 of the *SG13G2 Process Specification Rev. 1.2*, Copyright 2023 IHP PDK Authors, licensed under Apache-2.0, taken from IHP-Open-PDK and cropped out of the original page.

3.10 Technology Qualification

Before a product in a certain semiconductor technology is shipped to customers, a qualification procedure has to be completed. Its purpose is to demonstrate, on a statistically meaningful sample and before the first customer shipment, that the product will meet its datasheet over the intended lifetime and operating conditions. The overall framework is set by JEDEC

JESD47 (*Stress-Test-Driven Qualification of Integrated Circuits*), which references the individual test methods; automotive products additionally follow AEC-Q100, which defines temperature grades and larger sample sizes. Table 7 lists typical tests.

Table 7: Typical technology qualification tests

Test (examples)	Standard	Motivation & check
HTOL (high-temperature operating life)	JESD22-A108	Subject the IC to high temperature under operation over an extended duration.
ESD HBM	JS-001-2023	ESD test, human-body model.
ESD CDM	JS-002-2022	ESD test, charged-device model.
Latch-up	JESD78	Latch-up test.
Electrical test (ED)	JESD86	Check conformance to the datasheet across conditions and lots.

These tests fall into three groups: **lifetime tests** that accelerate wear-out, **robustness tests** that check survival of a single electrical stress event, and **electrical verification** against the datasheet.

HTOL stresses a sample of packaged devices—typically three lots of 77 parts, so that zero failures give a meaningful confidence bound—at an elevated junction temperature, usually 125 °C and often combined with a supply voltage above nominal, while the circuit is electrically exercised, for a duration of typically 1000 h. It accelerates exactly those wear-out mechanisms that limit the life of a MOSFET: **time-dependent dielectric breakdown (TDDB)** of the gate oxide, **hot-carrier injection (HCI)**, **bias temperature instability (BTI)**, and electro-migration in the interconnect. Since these mechanisms are thermally activated, an Arrhenius acceleration factor—commonly in the range of 20 to 100—translates the 1000 h of stress into years of operation at the application temperature. A device fails the test if it stops working, or if any parameter has drifted outside the datasheet limits.

ESD HBM models the electrostatic discharge of a charged person touching a pin. The **human-body model** discharges a 100 pF capacitor through a 1.5 kΩ resistor, which produces a comparatively slow pulse (rise time of a few nanoseconds, decay time constant of 150 ns) with a peak current of about 0.67 A per kV. Every pin is stressed against every supply, in both polarities; a common target is 2 kV. The protection consists of diodes at each pad plus a clamp across the supply, arranged so that there is a low-impedance path between *any* two pins—which is why ESD protection is a full-chip problem, not a per-pad one.

ESD CDM models the opposite situation: the device itself has been charged, for example by sliding through a shipping tube, and discharges when one of its pins touches a grounded surface. The **charged-device model** pulse is far faster than HBM, with a rise time below 500 ps and a total duration of about 1 ns, reaching peak currents of several amperes. Because the pulse is largely over before the protection structures have fully turned on, CDM is usually the harder of the two to pass, and it is the more relevant one in modern automated assembly, where a human rarely touches the parts. The stored charge grows with package size, so large

packages are stressed hardest. A typical target is 500 V, and interfaces between separate supply domains are the classic weak spot.

Latch-up targets the parasitic thyristor that is inherent to every CMOS well structure: the source of a PMOS, the n-well, the p-substrate, and the source of a neighboring NMOS form a PNPN structure, equivalent to a pair of cross-coupled parasitic bipolar transistors. If enough current is injected into a well—by an I/O pin driven beyond a supply rail, or by a supply overshoot—the structure latches into a low-impedance state that shorts supply to ground and persists until the power is removed, usually destroying the chip. JESD78 therefore injects ± 100 mA into the I/O pins and applies an overvoltage to the supplies, at the maximum operating temperature. The countermeasures are all layout measures: a generous number of well and substrate taps, which is why the flow above places one next to each transistor (Figure 37), guard rings around injecting circuits, and minimum-spacing rules between NMOS and PMOS.

Electrical test (ED) is not a stress test. It assesses the electrical parameters of devices from the manufacturing and test process being qualified, verifying that they meet every datasheet parameter across the specified supply-voltage and temperature range, and that they do so consistently across several production lots. This is what catches a systematic process shift that leaves the technology inside its own control limits, but the product outside its specification.

A complete qualification also covers the package rather than the silicon, with temperature cycling, highly accelerated stress testing (HAST) for moisture resistance, and solder-reflow sensitivity.

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